Marine Microbial Biodiversity

Lecture 06

Bacterial and Archaeal Diversity in Marine Environments

Diversity 1

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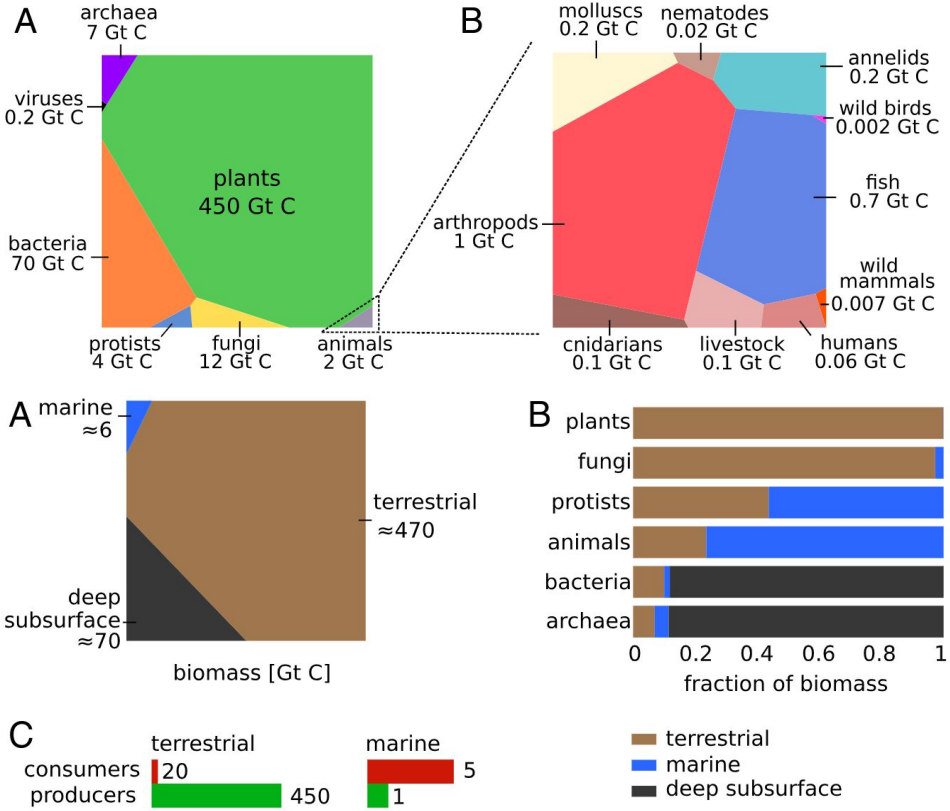


The Biomass Distribution On Earth

The sum of the biomass across all taxa on Earth is ~550 Gt C, of which bacteria constitute ~15% of the global biomass, as second most abundant component.

Although the ocean covers 71% of the Earth's surface, land biomass, at 470 Gt C, is about two orders of magnitude higher than the ~6 Gt C in marine biomass.

In marine environment, the most part is represented by animals and protists. Whereas bacteria and archaea cover the totality of the biomass distributed in the deep subsurface.

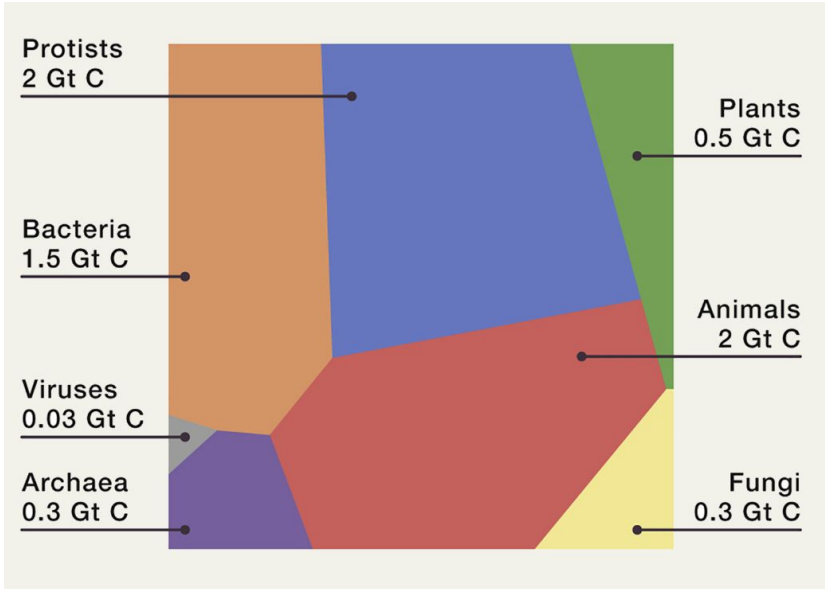


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10.1073/pnas.1711842115

The Biomass Distribution On Earth

In contrast to their domination on land, plants (green algae and seagrasses) account for less than 10% of the total biomass in the ocean (~6 Gt C), whereas animals correspond to ~33%. Viruses dominate the ocean in terms of number, with a virus-to-microbial cell ratio of 10:1, but constitute only ~0.5% of the total biomass. Marine microbes, including bacteria, archaea, protists, fungi and viruses, represent ~60% of total marine biomass, compared with ~18% on land.



These reported estimates for global marine biomass have been filtered to include only marine organisms. Therefore, all the microbial biomass, bacteria and archaea, living in the deep subsurface environment (deep aquifers and the ocean's crust) has not been even represented.

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The Biomass Distribution On Earth

Table 1.2 Distribution of microorganisms in and on Earth[a]

Habitat	*Percent of total*
Marine subsurface	66
Terrestrial subsurface	26
Surface soil	4.8
Oceans	2.2
All other habitats[b]	1.0

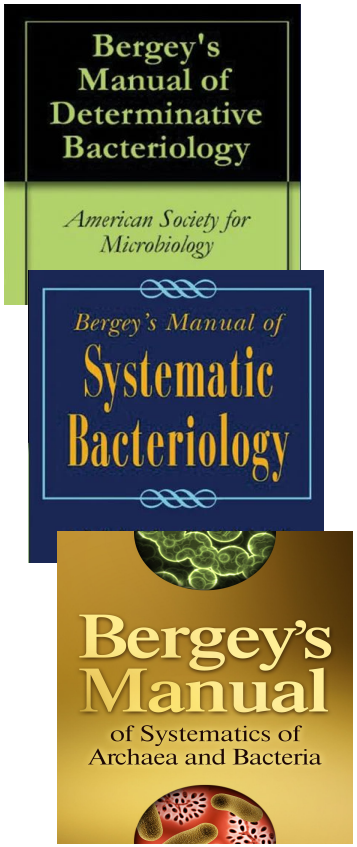
Most microbial cells reside in just a few very large habitats, and strange as it may seem, most do not reside on Earth's surface but instead lie underground in the oceanic and terrestrial subsurfaces at depths up to about 10 km. By comparison to the subsurface, surface soils and waters contain a relatively small percentage of the total microbial cells on Earth.

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The Roots Of Taxonomy In Microbiology

Taxonomy is the **identification**, **naming**, and **classification** of organisms. Early taxonomic systems had no theoretical basis; organisms were grouped according to apparent similarity.
Bergey's Manual of Determinative Bacteriology was first published in 1923 by David Hendricks Bergey, professor of bacteriology at the University of Pennsylvania. The manual's primary purpose was to provide a resource for the **identification** of bacteria based on **phenotypic characteristics**, such as **morphology**, **staining properties**, and **metabolic capabilities**.
A new multi-volume edition of Bergey's Manual of Systematic Bacteriology, which came out in four volumes from 1984 through 1989, attempted to organize bacterial species according to **known phylogenetic relationships**, an approach that continued with a second edition, published in five volumes from 2001 through 2012.



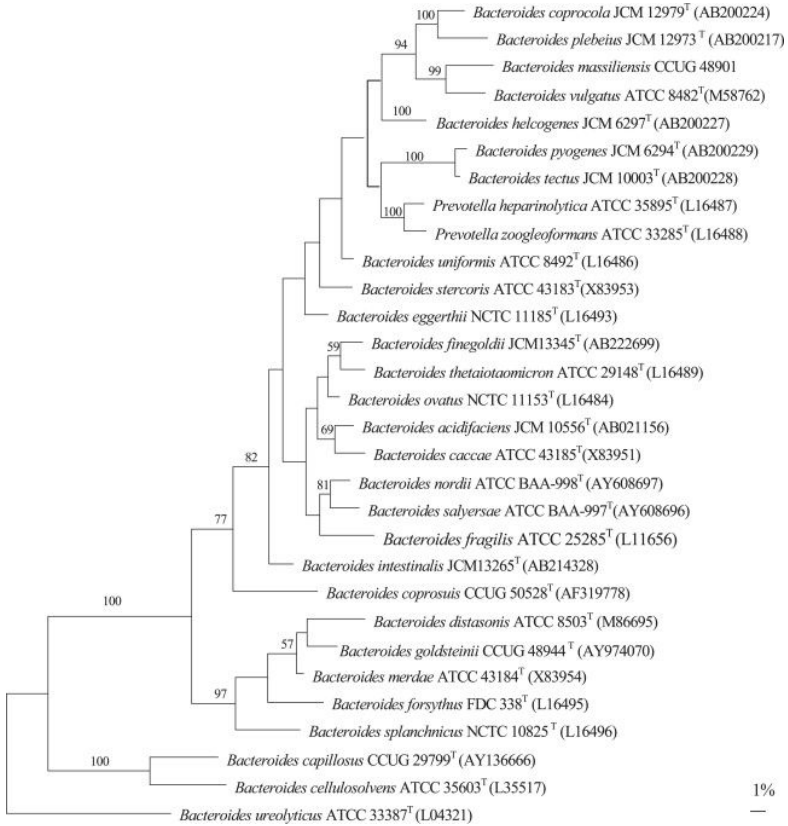
Characteristics	klebsiella pneumoniae
Oxygen requirement	
Motility	
Biochemistry:	
Oxidase activity	
Voges-Proskauer test	
Urea hydrolyzed	
Pectate hydrolyzed	
OPNG test	
Indole produced	
Malonate test	
Ornithine decarboxylated	
Lysinedecarboxylated	
Gluconate dehydrogenase	
Utilization of:	
D-Alanine	
L-Arabitol	
D-Galacturonate	
L-Rhamnose	
D-Saccharate	
D-Sorbitol	
L-Tyrosine	
Sucrose	
Glucose	
Fructose	
Type strain available:	
On what volume and pages of the Bergey's Manual did you found the descriptions of the bacteria?	

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The Roots Of Taxonomy In Microbiology

In 1930 was established the **International Committee on Systematics of Prokaryotes (ICSP)**. It is the body that oversees the nomenclature of prokaryotes and determines the rules by which prokaryotes are named and classified, first published in 1948 in the Code of Bacteriological Nomenclature now called **International Code of Nomenclature of Prokaryotes (ICNP)**. In 1950, a journal was established to disseminate the committee's conclusions to the microbiological community. It first appeared the following year under the title of International Bulletin of Bacteriological Nomenclature and Taxonomy, now known as the **International Journal of Systematic and Evolutionary Microbiology**.



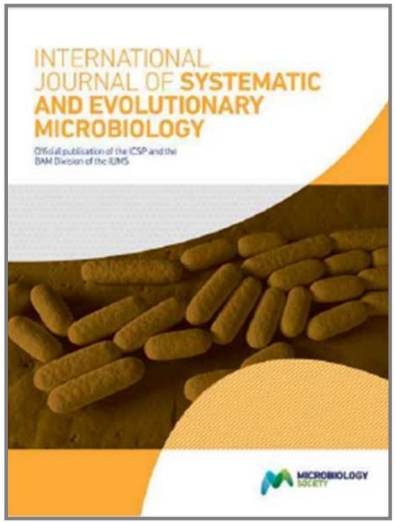
"This is the oath of a Knight of King Arthur's Round Table and should be for all of us to take to heart"

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The Roots Of Taxonomy In Microbiology

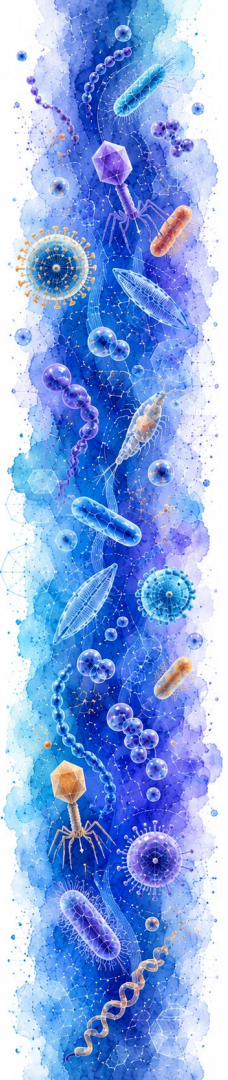
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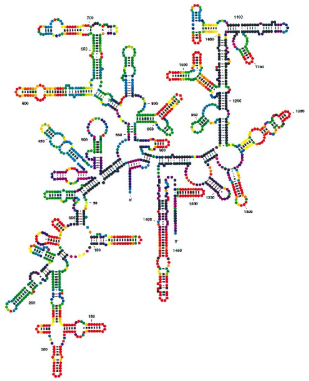
In the 1970s, the **ICSP** organized an extensive review that identified many inadequately described species as well as names that were redundant or inconsistent with established nomenclature. This culminated in the **Approved Lists of Bacterial Names**, which included validly published names for bacterial taxa and identified representative strains for all validly named and cultivable species, now become the online database **LPSN - List of Prokaryotic names with Standing in Nomenclature**, maintained by the Leibniz Institute DSMZ

LPSN .dsmz.de

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The Roots Of Taxonomy In Microbiology



Carl Woese's discovery of the **16S rRNA gene** as a molecular marker for bacterial classification in 1977 was a watershed moment. This led to the creation of new databases focusing on molecular data.

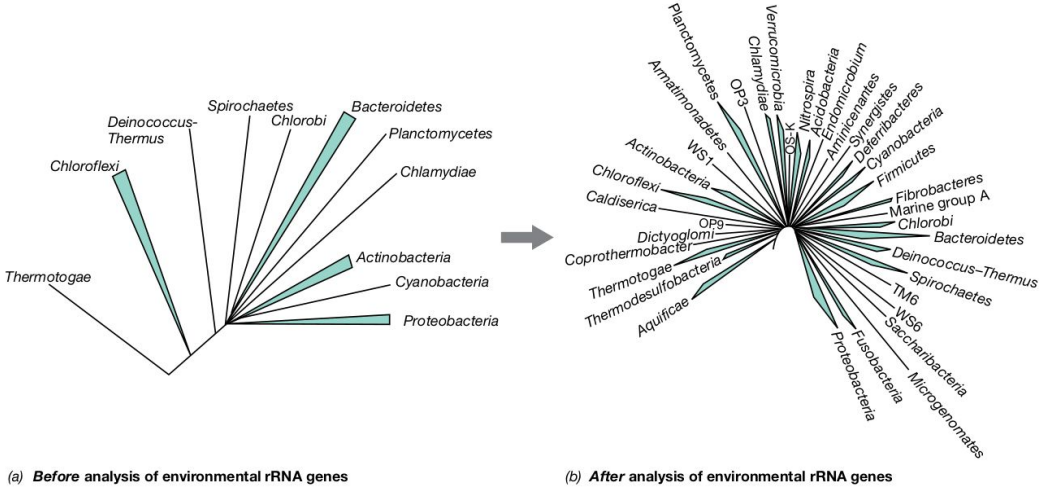
1980s: The Ribosomal Database Project (RDP) was one of the first specialized databases that emerged as sequencing technologies became available. It focused on 16S rRNA gene sequences, which could be used to classify bacteria and archaea in a more reliable, evolutionary context. RDP provides quality-controlled, aligned and annotated Bacterial and Archaeal 16S rRNA sequences, and Fungal 28S rRNA sequences, and a suite of analysis tools to the scientific community.

1990s: The National Center for Biotechnology Information (NCBI) Taxonomy database was launched in 1991, alongside GenBank. Its **main inclusion criterion** is the **availability of sequence data**: organisms must have genetic material, such as a genome or gene sequences, deposited in GenBank or related databases. If an organism lacks sequence data, it is excluded.
NCBI provided a **hierarchical classification** of all life forms, including microorganisms, and linked these classifications to genetic data, marking the **transition** from purely phenotypic classification to molecular and genomic classification.

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A New View Of The Tree Of Life

The use of increasingly new molecular techniques provide more accurate measurements of genetic identity and evolutionary relatedness, but results can be at odds with traditional classifications. The **cultivation-independent methods** of rRNA analysis from environmental samples pioneered by Norman Pace greatly improved our picture of microbial diversity.



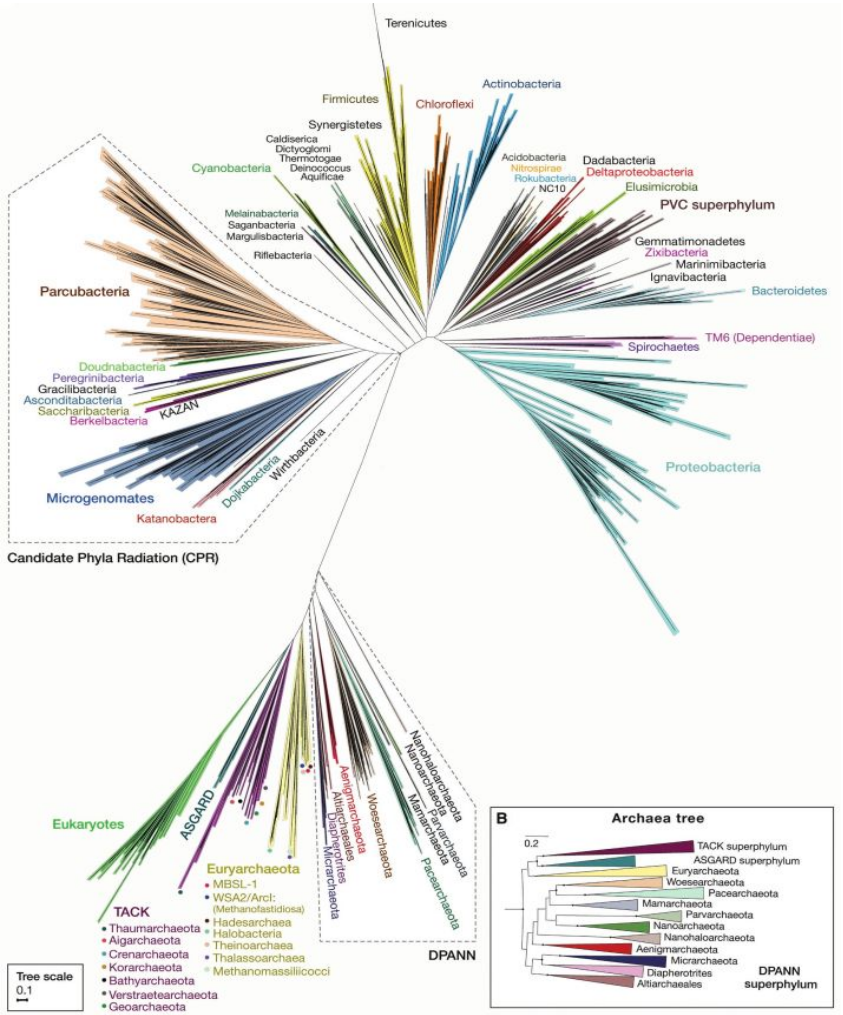
(a) ***Before*** **analysis of environmental rRNA genes** (b) ***After*** **analysis of environmental rRNA genes**

Analysis of environmental rRNA genes leads to **discovery of new phyla** of Bacteria. a) In 1987 Carl Woese described 11 phyla of Bacteria from analysis of rRNA genes from cultured species. b) By 1998, analyses of rRNA genes from environmental samples, as described by Norman Pace, had revealed evidence for 36 bacterial phyla.

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A New View Of The Tree Of Life

Total diversity is represented by sequenced genomes, highlighting the diversity of major accepted bacterial lineages, on the right. On the left, the **Candidate Phyla Radiation (CPR)**, which is composed of ultra-small bacteria have been **difficult to study using traditional cultivation techniques**, and are still in the process of definition at lower taxonomic levels. The lower section shows the eukaryotes as a single cluster and the Archaea as branches representing the phyla and newly designated super-phyla.

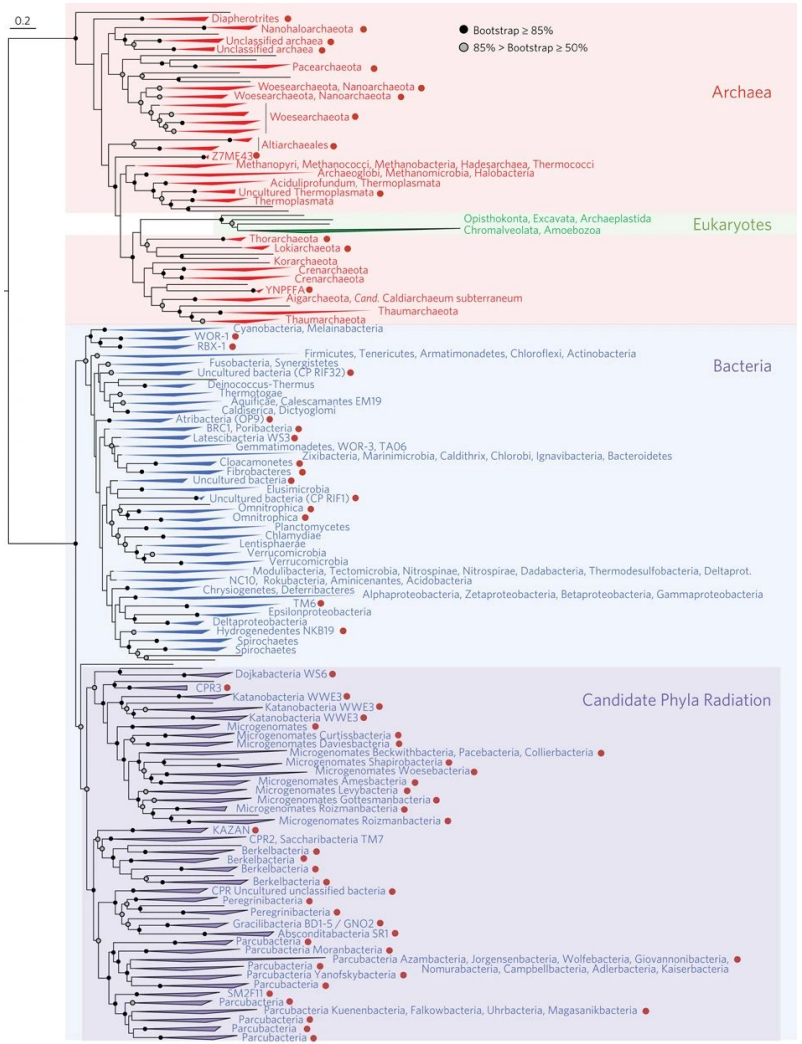


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A New View Of The Tree Of Life

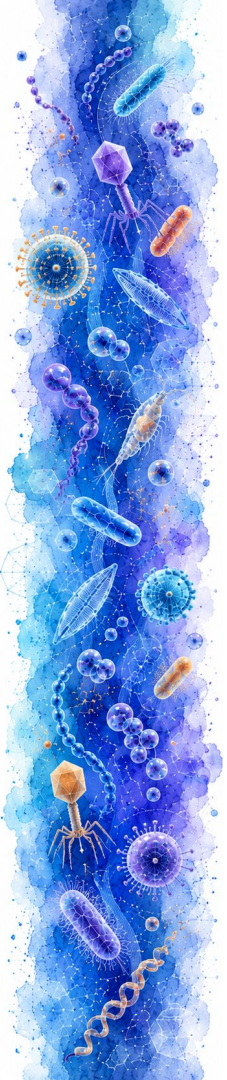
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Regardless of branching order, the CPR, in combination with **other taxa in bacterial and archaeal domains that lack isolated representatives**, clearly comprises the **majority of life's current diversity**.



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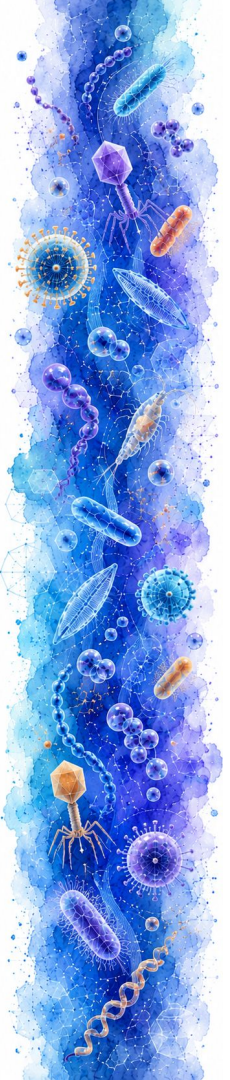
International Code Of Nomenclature Of Prokaryotes - 2022 Revision

To date, the International Code of Nomenclature of Prokaryotes recognizes only cultures as 'type material', thus uncultured organisms cannot be formally named.

"Rule 18a: **The type strain is made up of living cultures of an organism, which are descended from a strain designated as the nomenclatural type**. The strain should have been maintained in pure culture and its characters should agree closely to its characters with those in the original description (see Chapter 4C). The type strain may be designated in various ways (see Rules 18b, 18c, and 18d).
As from 1 January 2001, **no further descriptions, preserved (non-viable) specimens, or illustrations may be designated as the type**.
Rule 30: For the name of a species to be validly published, it must conform to the following conditions.
As of 1 January 2001, the valid publication of the name of a new species, or a new combination previously represented by a viable culture must include the designation of a type strain (see Rule 18a), and **a viable culture of that strain must be deposited in at least two publicly accessible culture collections** in different countries from which subcultures must be available."

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10.1099/ijsem.0.005585



The Tree Of Life: Bacteria and Archaea

LPSN .dsmz.de

List of Prokaryotic names with Standing in Nomenclature (LPSN) is an online database that maintains information on the **naming and taxonomy** of prokaryotes, following the taxonomy requirements and rulings of the ICNP. Here listed 119 entries for the bacterial domain, of which 39 are accepted phyla with accepted names, 79 are only proposed and 1 includes several bacteria not officially assigned to a taxon.

Abditibacteriota	"*Candidatus* Binatota"	***Cyanobacteriota***	"*Candidatus* Fervidibacterota"	"*Candidatus* Kryptoniota"	"*Candidatus* Neomarinimicrobiota"	"*Candidatus* Rifleibacteriota"
"*Candidatus* Acetithermota"	"*Candidatus* Bipolaricaulota"	***Deferribacterota***	***Fibrobacterota***	"*Candidatus* Latescibacteria"	"*Candidatus* Nitrosediminicolota"	"*Candidatus* Saccharibacteria"
Acidobacteriota	"*Candidatus* Caldipriscota"	"*Candidatus* Deferrimicrobiota"	***Fusobacteriota***	"*Candidatus* Latescibacterota"	***Nitrospinota***	"*Candidatus* Saccharimonadota"
Actinomycetota	***Caldisericota***	***Deinococcota***	***Gemmatimonadota***	***Lentisphaerota***	***Nitrospirota***	***Spirochaetota***
"*Candidatus* Aerophobetes"	***Calditrichota***	"*Candidatus* Dependentiae"	"*Candidatus* Goldbacteria"	"*Candidatus* Lernaellota"	"*Candidatus* Omnitrophica"	"*Candidatus* Sumerlaeota"
"*Candidatus* Aerophobota"	"*Candidatus* Calescamantes"	***Dictyoglomerota***	"*Candidatus* Goldiibacteriota"	"*Candidatus* Lithacetigenota"	"*Candidatus* Omnitrophota"	***Synergistota***
"*Candidatus* Altimarinota"	"*Candidatus* Calescibacteriota"	*Dictyoglomota*	"*Candidatus* Gracilibacteria"	"*Candidatus* Macinerneyibacteriota"	"*Candidatus* Pacebacteria"	"*Candidatus* Tectimicrobiota"
Aquificota	"*Candidatus* Canglongiota"	"*Candidatus* Dormibacteraeota"	"*Candidatus* Heilongiota"	"*Candidatus* Margulisbacteria"	"*Candidatus* Paceibacterota"	"*Candidatus* Tectomicrobia"
Armatimonadota	***Chlamydiota***	"*Candidatus* Dormiibacterota"	"*Candidatus* Hinthialibacterota"	"*Candidatus* Margulisiibacteriota"	"*Candidatus* Parcunitrobacterota"	***Thermodesulfobacteriota***
"*Candidatus* Atribacteria"	***Chlorobiota***	"*Candidatus* Effluviviacota"	"*Candidatus* Hydrogenedentes"	"*Candidatus* Methylomirabilota"	"*Candidatus* Peregrinibacteria"	***Thermomicrobiota***
Atribacterota	***Chloroflexota***	"*Candidatus* Effluvivivacota"	"*Candidatus* Hydrogenedentota"	"*Candidatus* Microgenomates"	"*Candidatus* Peregrinibacteriota"	***Thermotogota***
"*Candidatus* Auribacterota"	***Chrysiogenota***	"*Candidatus* Electryoneota"	"*Candidatus* Hydrothermae"	"*Candidatus* Microgenomatota"	***Planctomycetota***	"*Candidatus* Tianyaibacteriota"
"*Candidatus* Babelota"	"*Candidatus* Cloacimonadota"	"*Candidatus* Elulimicrobiota"	"*Candidatus* Hydrothermota"	"*Candidatus* Modulibacteria"	***Pseudomonadota***	***Verrucomicrobiota***
Bacillota	"*Candidatus* Cloacimonetes"	"*Candidatus* Elulota"	"*Candidatus* Kapaibacteriota"	"*Candidatus* Moduliflexota"	"*Candidatus* Pyropristinus"	***Vulcanimicrobiota***
[Bacteria, not assigned to phylum]	***Coprothermobacterota***	***Elusimicrobiota***	***Kiritimatiellota***	"*Candidatus* Muirbacteria"	"*Candidatus* Qinglongiota"	"*Candidatus* Wirthbacteria"
Bacteroidota	"*Candidatus* Cosmopoliota"	"*Candidatus* Fermentibacterota"	"*Candidatus* Krumholzibacteriota"	"*Candidatus* Muirbacteriota"	***Rhodothermota***	"*Candidatus* Wirthibacterota"
Balneolota	"*Candidatus* Cryosericota"	"*Candidatus* Fervidibacteria"	"*Candidatus* Kryptonia"	***Mycoplasmatota***	"*Candidatus* Riflebacteria"	"*Candidatus* Zhurongbacterota"

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https://lpsn.dsmz.de/domain

The Tree Of Life: Bacteria And Archaea

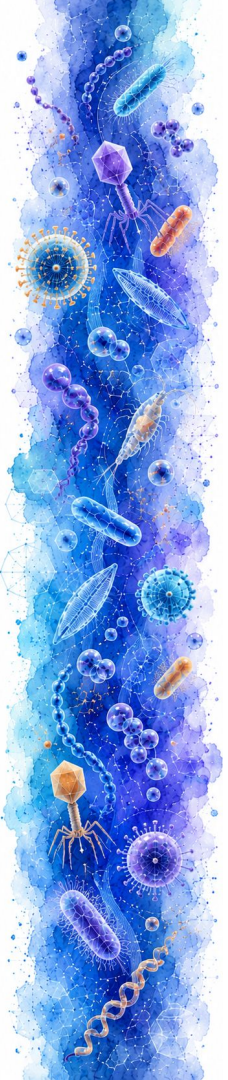
List of Prokaryotic names with Standing in Nomenclature (LPSN) is an online database that maintains information on the naming and taxonomy of prokaryotes, following the taxonomy requirements and rulings of the ICNP. For the archaeal domain are listed 28 entries, of which only 5 are accepted phyla with accepted names, 22 only proposed, and 1 includes several archaea not officially assigned to a taxon.

Archaea Hide phylum list

- "*Candidatus* Aenigmatarchaeota"
- "*Candidatus* Aigarchaeota"
- [Archaea, not assigned to phylum]
- "*Candidatus* Augarchaeota"
- "*Candidatus* Diapherotrites"
- "*Candidatus* Freyrarchaeota"
- "*Candidatus* Geoarchaeota"
- "*Candidatus* Hadarchaeota"
- "*Candidatus* Hadesarchaeota"
- "*Candidatus* Hodarchaeota"
- "*Candidatus* Huberarchaea"
- "*Candidatus* Huberarchaeota"
- "*Candidatus* Hydrothermarchaeota"
- "*Candidatus* Iainarchaeota"
- "*Candidatus* Kariarchaeota"
- ***Methanobacteriota***
- "*Candidatus* Micrarchaeota"
- ***Microcaldota***
- ***Nanobdellota***
- "*Candidatus* Nanohaloarchaeota"
- "*Candidatus* Nezhaarchaeota"
- "*Candidatus* Parvarchaeota"
- "*Candidatus* Poseidoniota"
- ***Promethearchaeota***
- "*Candidatus* Sigynarchaeota"
- "*Candidatus* Thermoplasmatota"
- ***Thermoproteota***
- "*Candidatus* Undinarchaeota"

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The Tree Of Life: Bacteria and Archaea

As the number of environmental small subunit (SSU) ribosomal RNA gene sequences has greatly surpassed the number of cultured microorganisms, reconciliation of the established taxonomy and classification of the uncultured microorganisms are crucial. Recently, rational **taxonomic boundaries** (thresholds) have been proposed for high taxa of bacteria and archaea on the basis of 16S rRNA gene sequence identities.

	LPSN 2012		SILVA REF 114 incremental‡		SILVA REF 114 single-step‡	SILVA REF 114 ambiguous§
	Bacteria	Archaea	Bacteria	Archaea	Bacteria and archaea	Bacteria and archaea
Sequences			1,265,442	41,228	1,306,670	31,469
Species	9,624	391	246,841	13,159	241,254	28,983
Genera	1,916	113	90,068	4,383	80,939	19,607
Families	290	27	16,719	1,239	14,369	4,803
Orders	122	15	6,230	682	5,366	1,148
Classes	79	9	2,969	431	2,573	408
Phyla	27	2	1,481	287	1,356	84

98.7% (species)
94.5% (genus)
86.5% (family)
82.0% (order)
78.5% (class)
75.0% (phylum)

All sequences that had a quality score of less than 75% were reported as ambiguous taxa

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10.1038/nrmicro3330

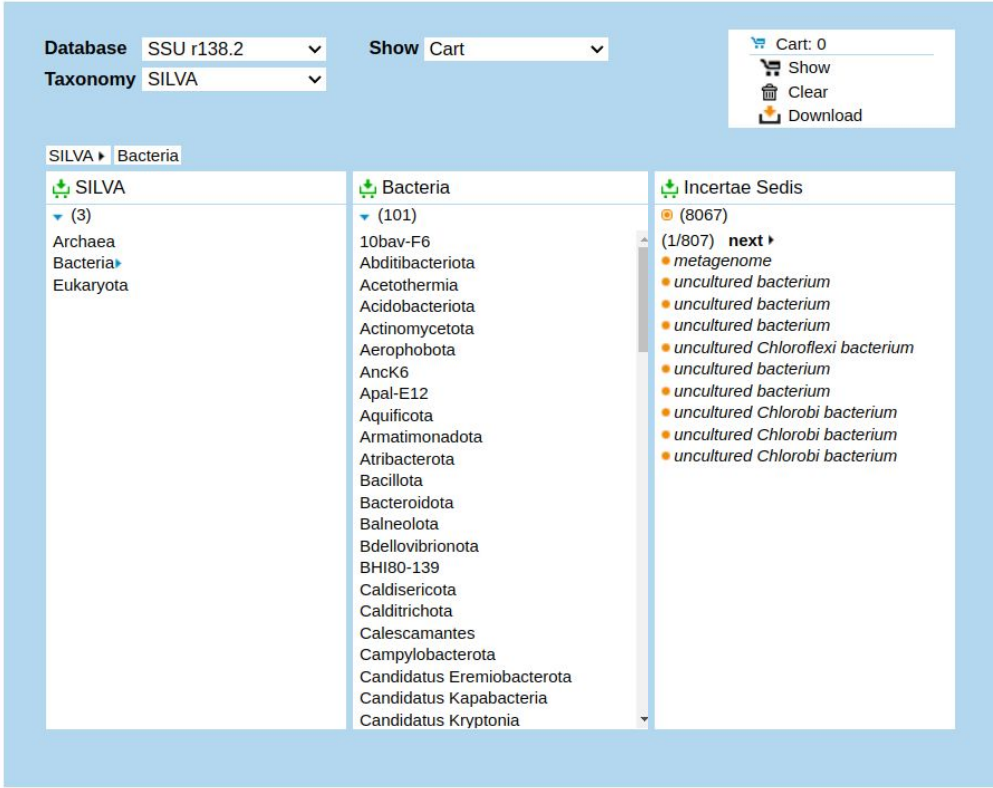
The Tree Of Life: Bacteria And Archaea

SILVA is a high quality **ribosomal RNA database**. Represent a comprehensive online resource for quality checked and aligned ribosomal RNA sequence data.

SILVA SSU 138.2 update release

	SSU Parc	SSU Ref NR 99	LSU Parc	LSU Ref NR 99
Minimal length	300	1200/900	300	1900
Quality filtering	basic	strong	basic	strong
Guide Tree	no	yes	no	yes
Release date	11.07.24	11.07.24	11.07.24	11.07.24
Aligned rRNA sequences	**9,469,070**	510,495	**1,312,521**	95,279

For the **bacterial domain**, here listed 8521 entries, of which 100 indicate recognised or candidate phyla and 8421 Incertae sedis representing sequences obtained from uncultured, unidentified or metagenome-derived organisms.



https://www.arb-silva.de/browser/

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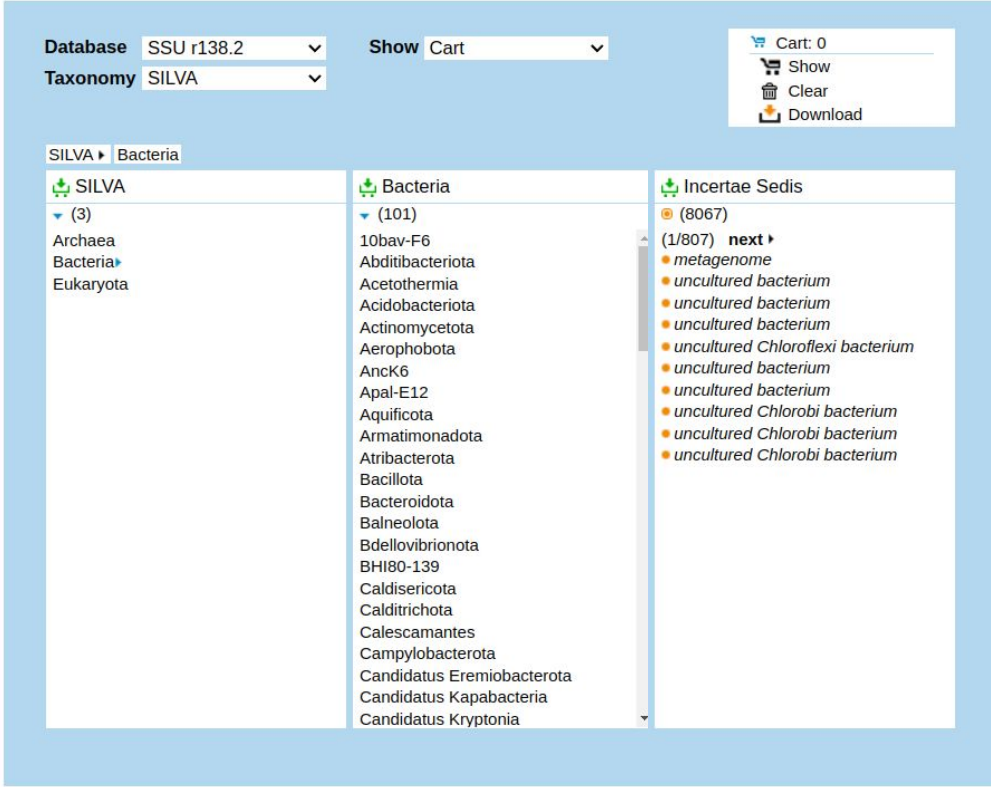
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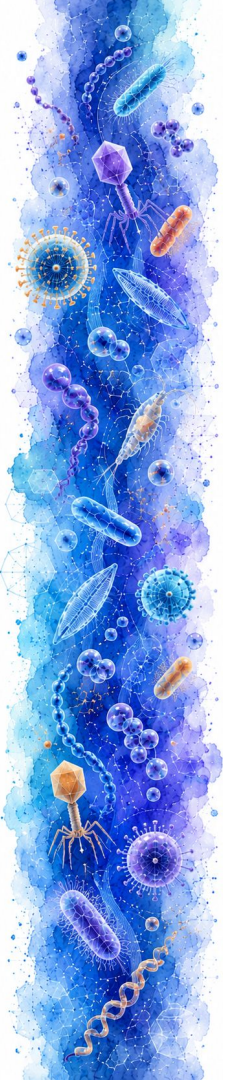
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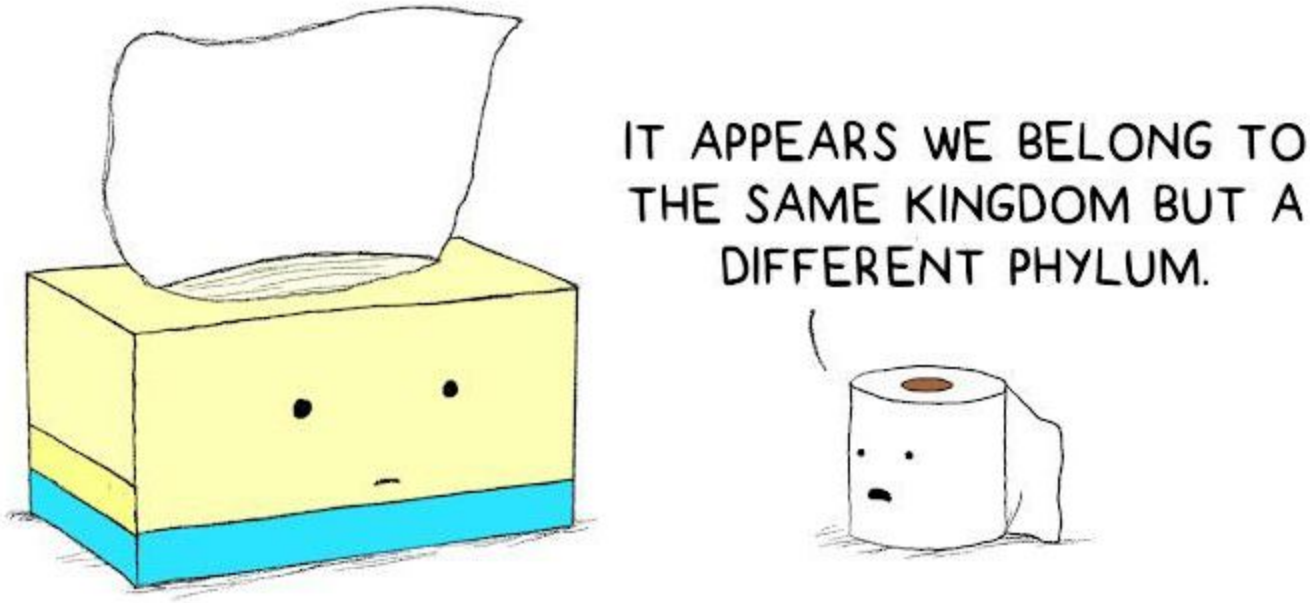
For the **archaeal domain**, here listed 87 entries, of which 17 indicate recognised or candidate phyla and 70 Incertae sedis representing sequences obtained from uncultured, unidentified or metagenome-derived organisms.



https://www.arb-silva.de/browser/

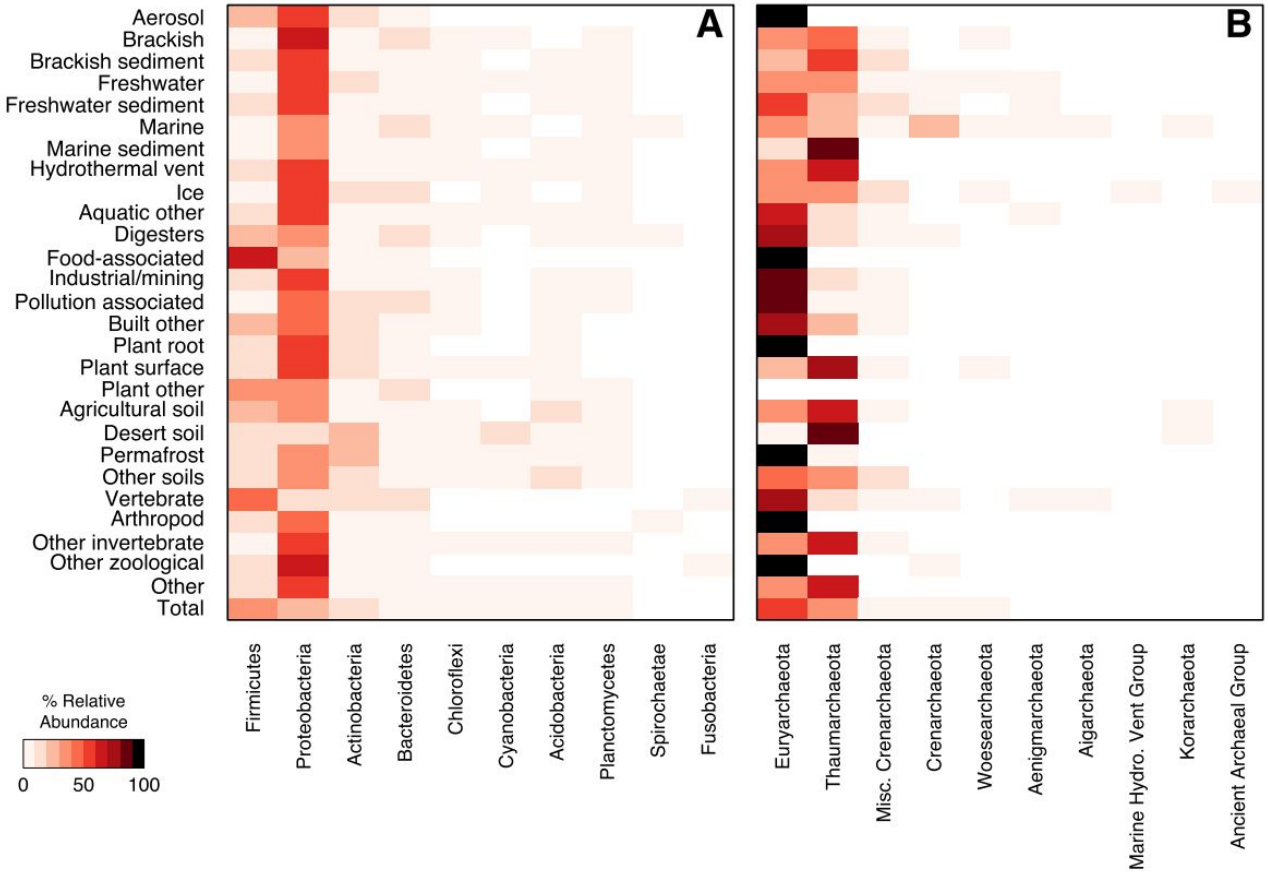
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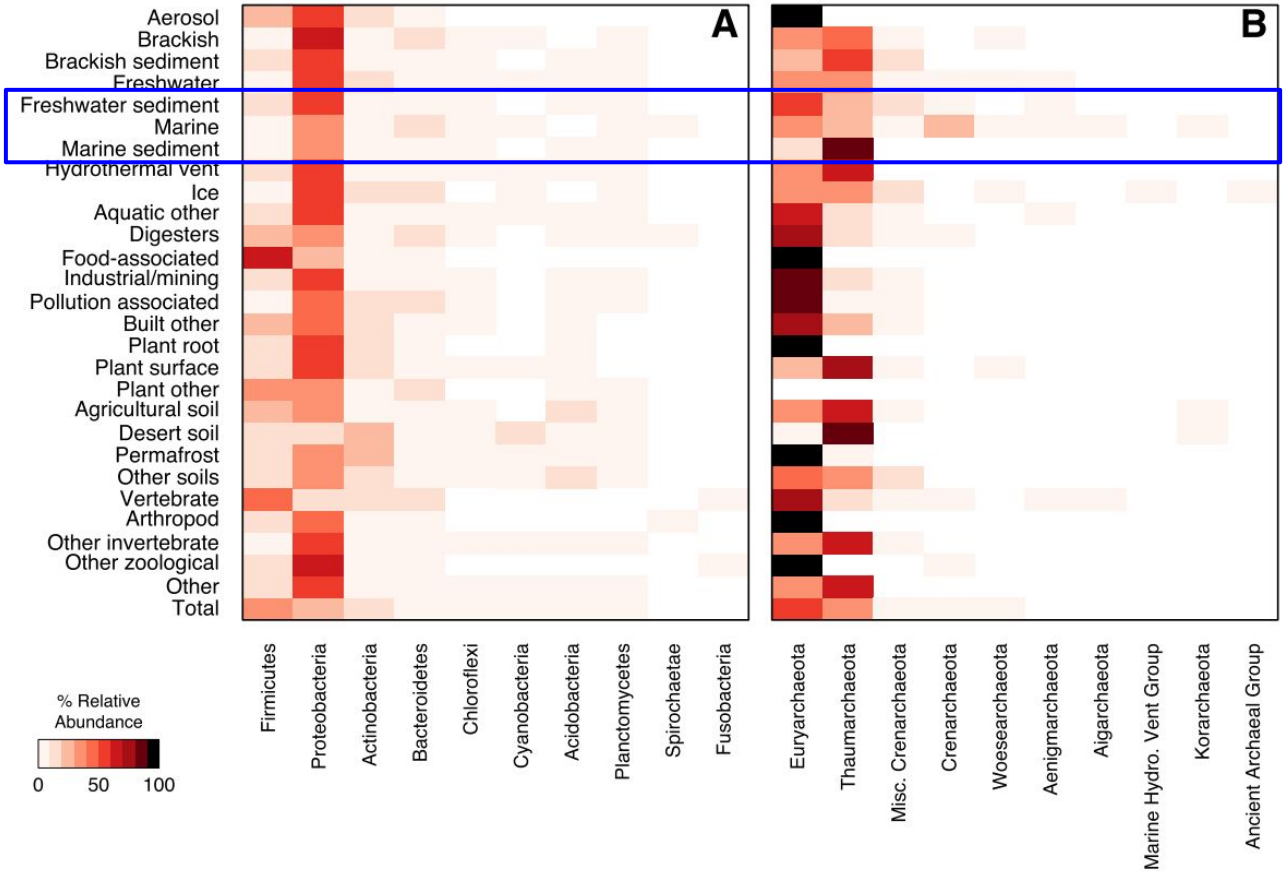
The Tree Of Life: Bacteria And Archaea



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doi: 10.1128/mBio.00201-16

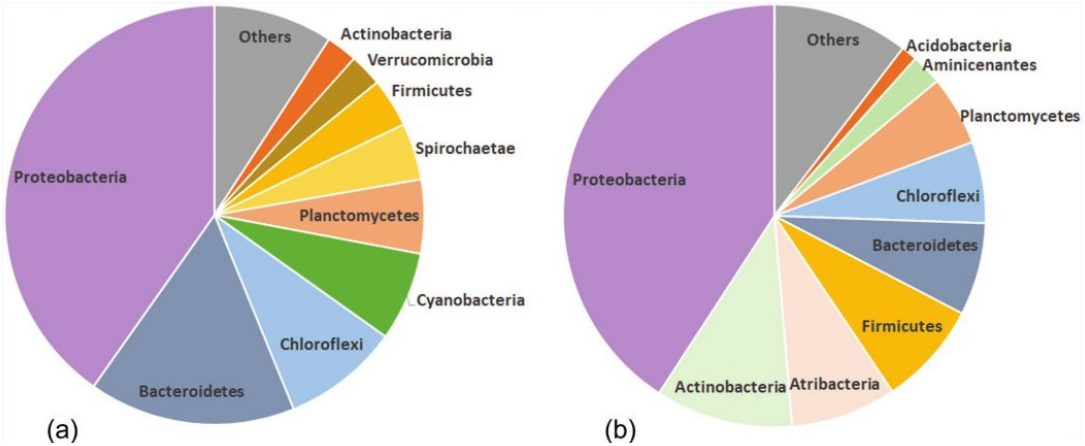
The Tree Of Life: Bacteria And Archaea



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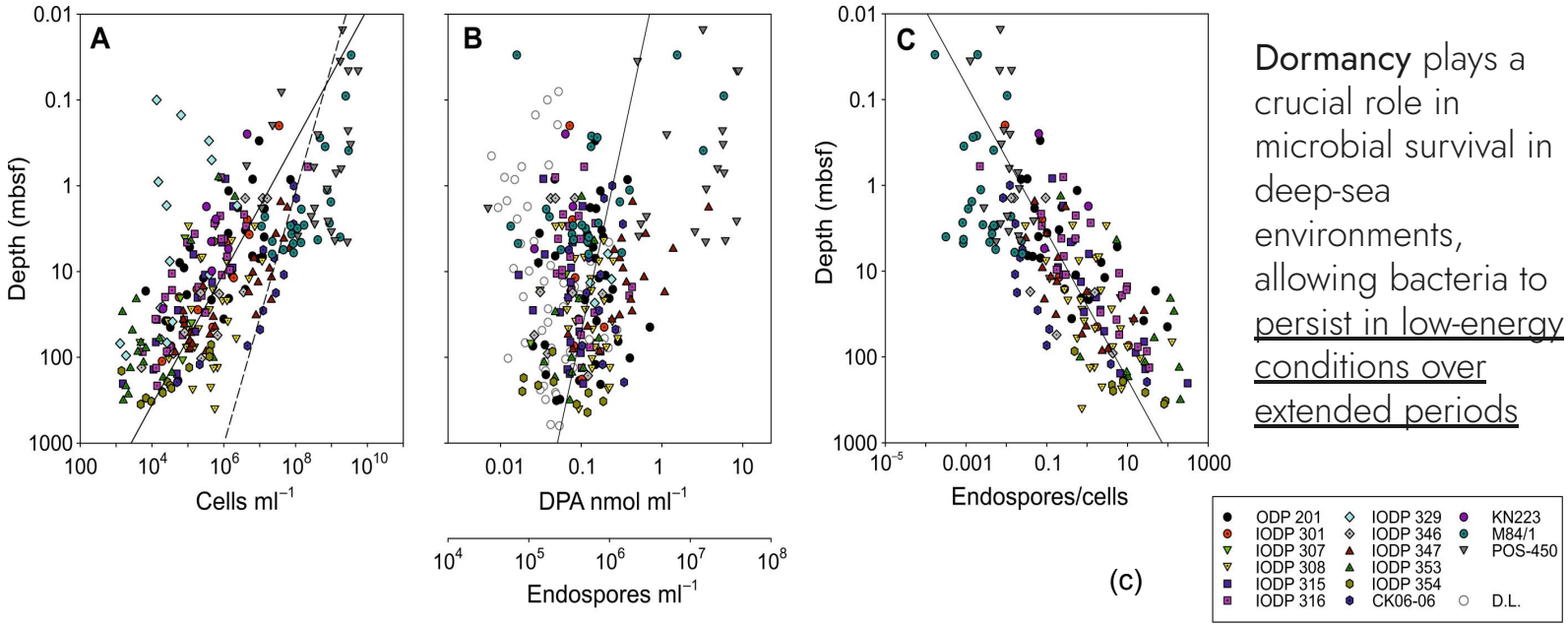
The Tree Of Life: Bacteria and Archaea (In Marine Environment)



Frequency of sequences for bacterial phyla from **marine water samples (a)** and **marine sediments and hydrothermal vents (b)**, and for **archaeal phyla (c)**.

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The Tree Of Life: Bacteria and Archaea (In Marine Environment)



Dormancy plays a crucial role in microbial survival in deep-sea environments, allowing bacteria to persist in low-energy conditions over extended periods

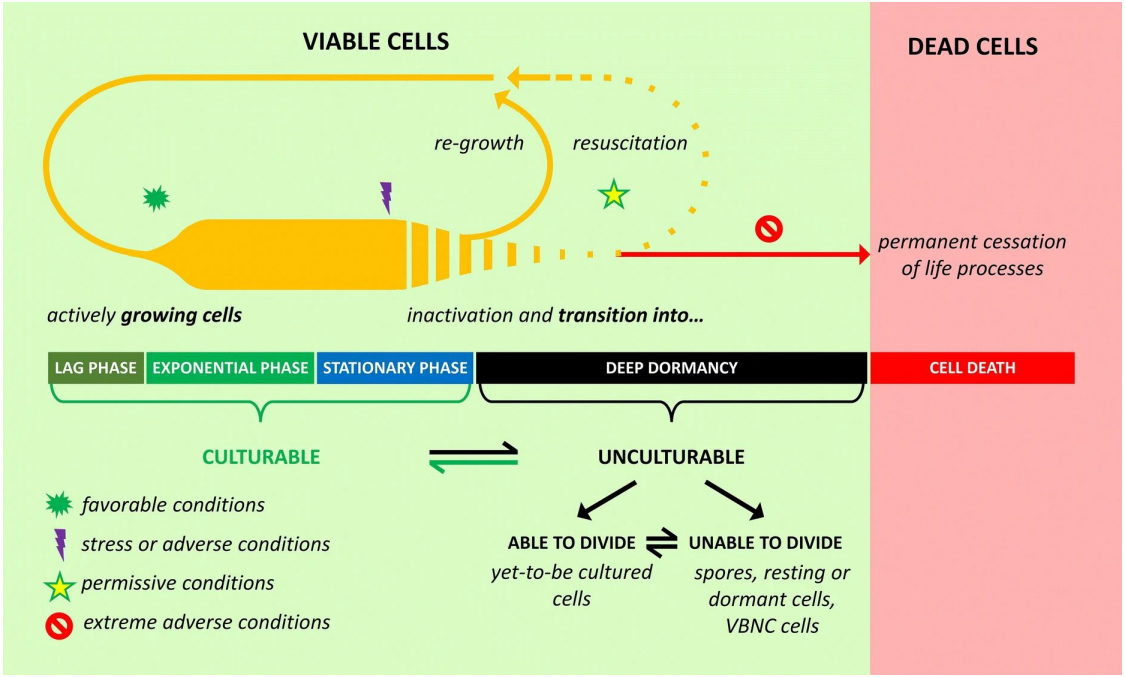
Distribution of vegetative cells and endospores versus depth. Concentrations of (A) vegetative cells and (B) endospores and (C) ratio of endospores to vegetative cells versus sediment depth (meters below the seafloor) in samples from various Ocean Drilling Program (ODP) and Integrated Ocean Drilling Program (IODP) expeditions and national research cruises.

DOI: 10.1126/sciadv.aav1024

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Different Physiological States In Microbes

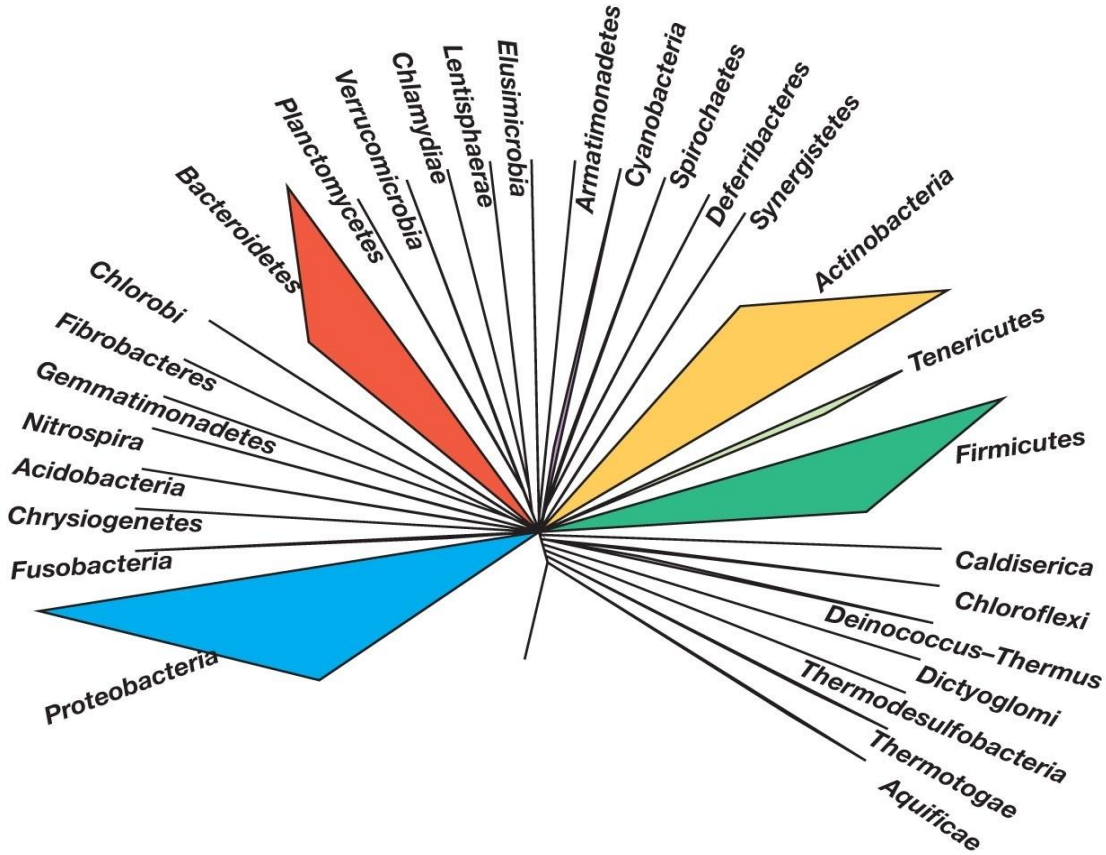
Viable But NonCulturable (VBNC) microbes refers as to microbes that are in a state of very low metabolic activity and do not divide, but are alive and have the ability to become culturable once resuscitated.
Various factors influence the culturability of these microorganisms, such as nutrient availability and signaling metabolites.



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10.1007/s11157-020-09522-4

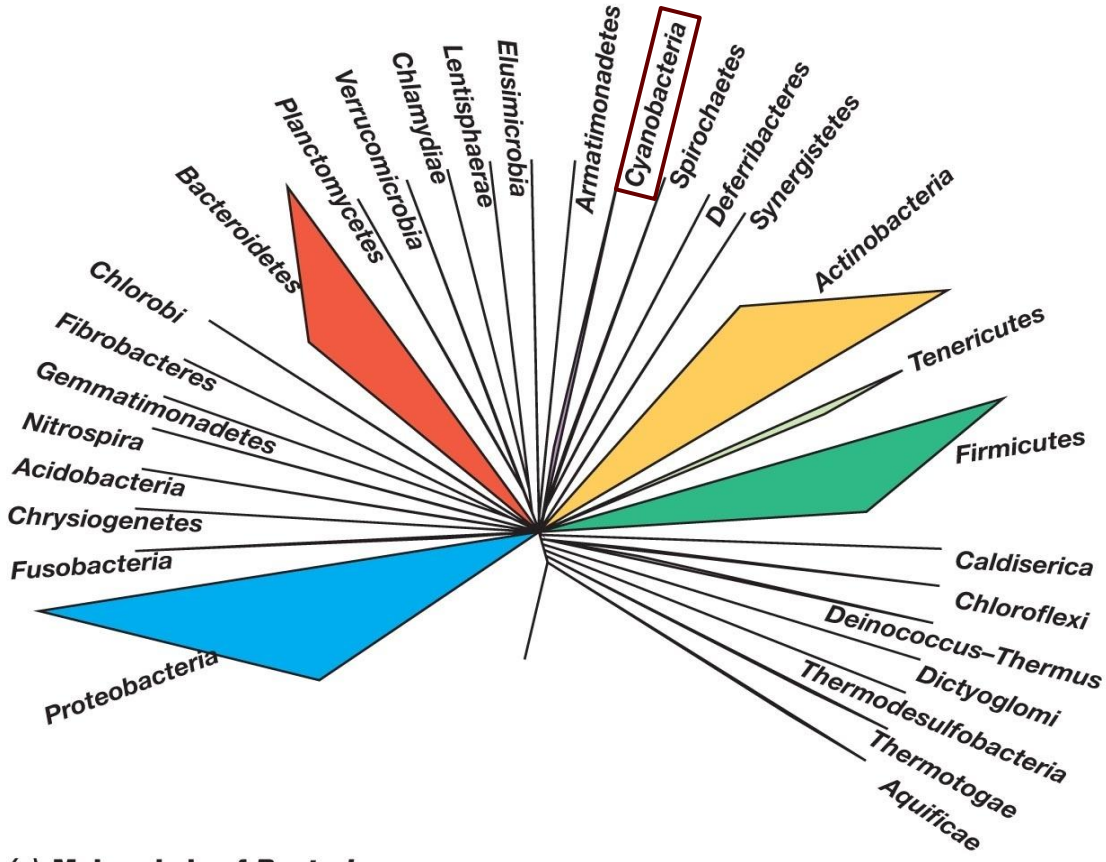
The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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The Tree Of Life: Bacteria

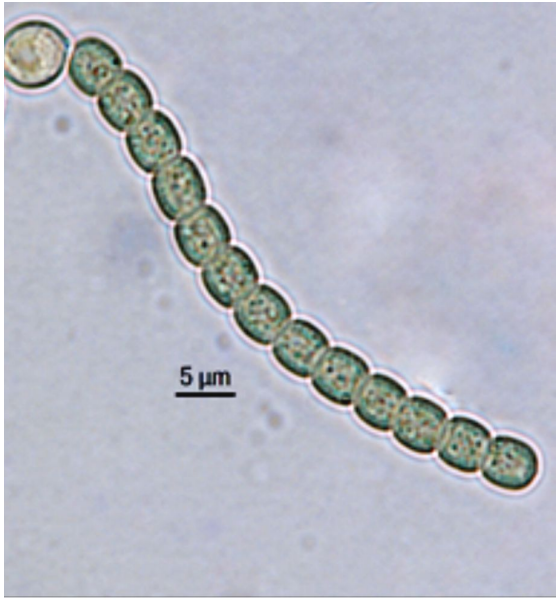


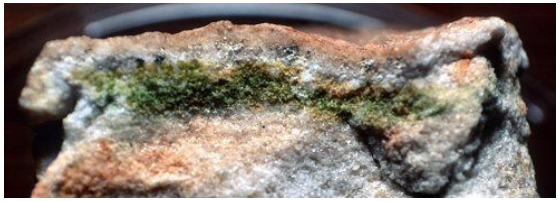
(a) **Major phyla of *Bacteria***

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Bacterial phyla: Cyanobacteria

Cyanobacteria (once known as Cyanophyta), are a phylum of **photosynthetic**, **carbon-fixing**, in some cases also **nitrogen-fixing**, bacteria. Cyanobacteria can be found in almost every terrestrial and aquatic habitat - oceans, fresh water, damp soil, temporarily moistened rocks in deserts, bare rock and soil, and even Antarctic rocks. They can occur as **planktonic cells** or form phototrophic **biofilms**. They are found in almost every **endolithic ecosystem** and can form **symbiosis** (cyanobionts).
The morphological diversity of the Cyanobacteria is huge. Cells range in size from 0.5 μm to 100 μm in diameter. They show from unicellular to filamentous cells organization, including colonial species, forming filaments, sheets, or even bunches. In filamentous species cells can differentiate into several different cell types: vegetative cells, nitrogen-fixing heterocysts, spore-like akinetes.
Photosynthesis is performed in distinctive folds in the outer membrane of the cell, called thylakoids. Chloroplasts found in eukaryotes, have their ancestry in Cyanobacteria, via a process called endosymbiosis.





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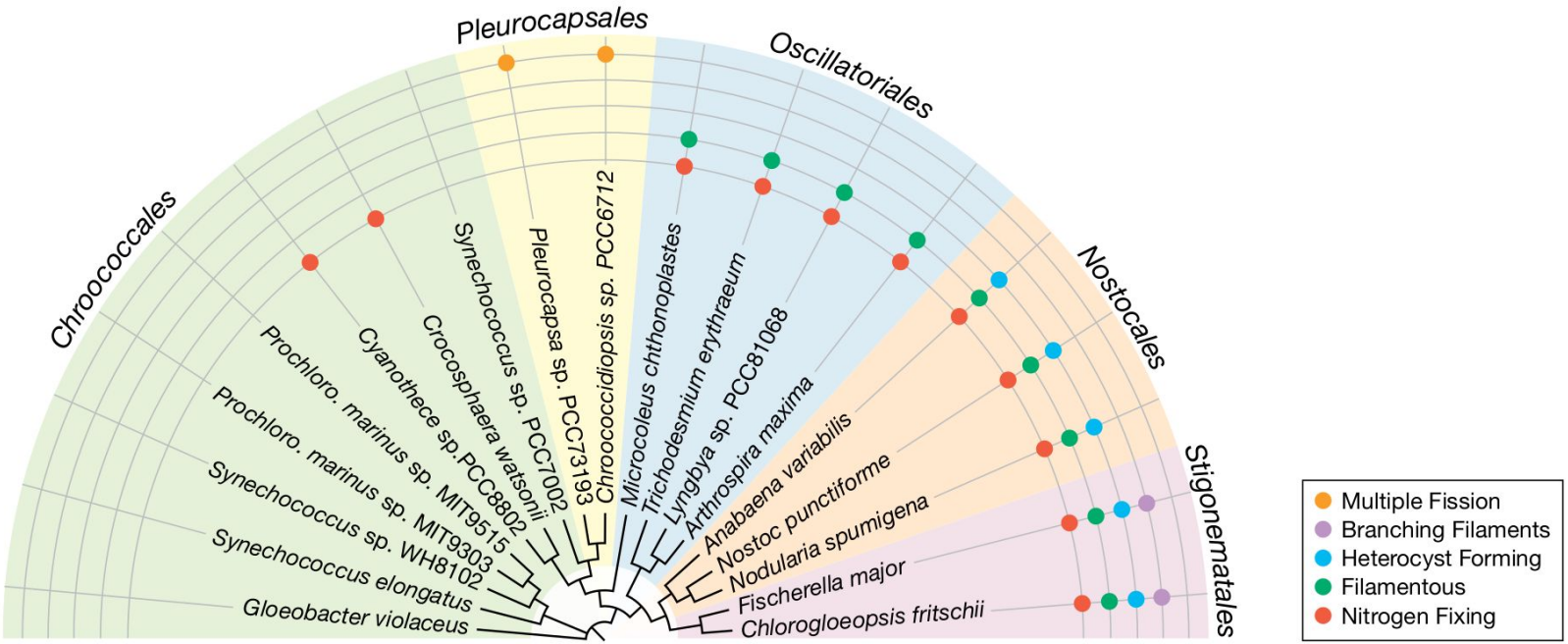
Bacterial Phyla: Cyanobacteria

Cyanobacteria are **oxygenic phototrophs**, having both type I and type II photosystems. All species fix CO_2 by the Calvin cycle, and many can fix N_2.
Some Cyanobacteria can be **photoheterotrophs**, using simple organic compounds as both carbon and electron source when light is present. A few cyanobacteria, mainly filamentous species, are **mixotrophic**, growing in the dark on glucose or sucrose, using the sugar as both carbon and energy source.
When **sulfide** concentrations are **high**, some cyanobacteria are able to **switch** from oxygenic photosynthesis **to anoxygenic photosynthesis** using hydrogen sulfide rather than water as electron donor.
Cyanobacteria are of central importance to the productivity of the oceans. Small unicellular cyanobacteria of the genera **Synechococcus** and **Prochlorococcus** are the most abundant phototrophs in the oceans. Together these organisms contribute ~80% of marine photosynthesis and ~35% of all photosynthetic activity on Earth.
Cyanobacteria are thought to have been the main contributor to the Great Oxygenation Event, ca. 2.5 Ga



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Bacterial Phyla: Cyanobacteria



Taxonomically informative traits mapped onto the phylogeny of Cyanobacteria (main orders)

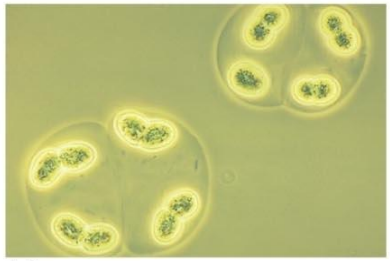
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Bacterial Phyla: Cyanobacteria

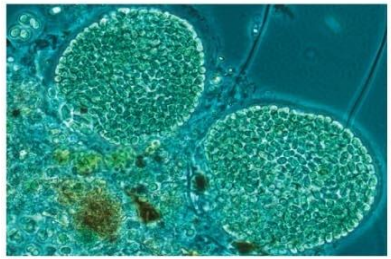
Table 14.1 Genera and grouping of cyanobacteria

Group	Genera
Group I, *Chroococcales.* Unicellular or cell aggregates	*Gloeothece* (Figure 14.2*a*), *Gloeobacter, Synechococcus, Cyanothece, Gloeocapsa, Synechocystis, Chamaesiphon, Merismopedia, Crocosphaera* (Figure 14.7*a*), *Prochlorococcus, Prochloron*
Group II, *Pleurocapsales.* Reproduce by formation of small spherical cells called baeocytes produced through multiple fission	*Pleurocapsa* (Figure 14.2*b*), *Dermocarpa, Xenococcus, Dermocarpella, Myxosarcina, Chroococcidiopsis*
Group III, *Oscillatoriales.* Undifferentiated filamentous cells that divide by binary fission in a single plane	*Lyngbya* (Figure 14.2*c*), *Spirulina* (Figure 14.5), *Arthrospira, Oscillatoria* (Figure 14.6*a–b*), *Microcoleus, Pseudanabaena, Trichodesmium* (Figure 14.7*b*)
Group IV, *Nostocales.* Filamentous cells that produce heterocysts	*Nodularia* (Figures 14.2*d*), *Nostoc, Calothrix* (Figure 14.8*a–b*), *Anabaena* (Figure 14.6*c*), *Cylindrospermum, Scytonema, Richelia* (Figure 14.7*c*)
Group V, *Stigonematales.* Cells divide to form branches	*Fischerella* (Figures 14.2*e*, 14.8*c, d*), *Stigonema, Chlorogloeopsis, Hapalosiphon*

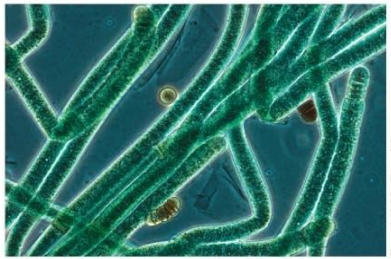
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Susan Barns and Norman Pace

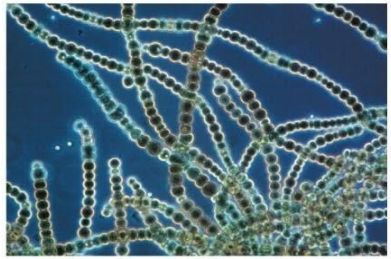
(a)


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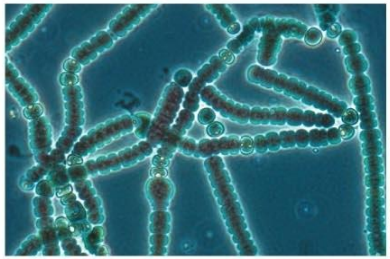
(b)


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(c)


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(d)

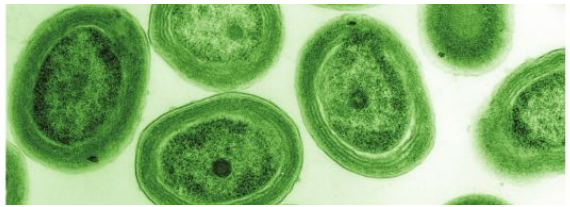

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(e)

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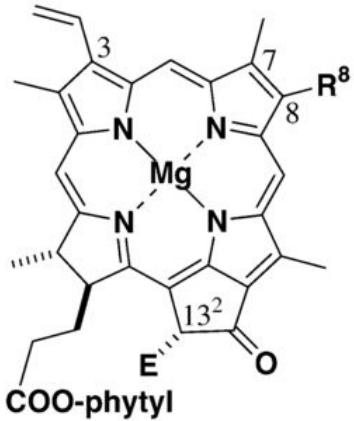
Cyanobacteria: *Prochlorococcus marinus*

Prochlorococcus is a genus of very small (0.6 μm) marine cyanobacteria with an **unusual pigmentation**, due to the synthesis of modified forms of chlorophyll (**divinyl chlorophylls** a and b) and the absence of phycobilins.

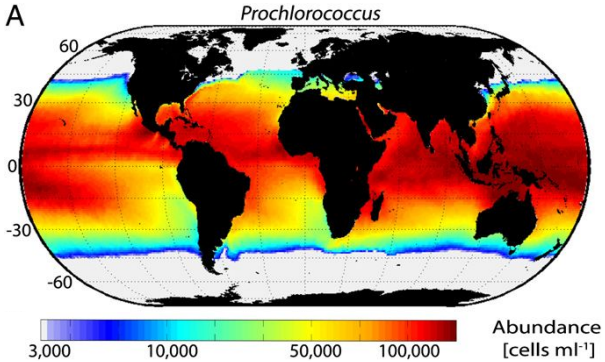


Prochlorococcus is the dominant phototroph in **oligotrophic** ocean waters. It inhabits large parts of the euphotic zone of the world's tropical oceans, at depths of 25 - 200 m, and at temperatures ranging from 10 to 33 °C, with some strains able to grow at depths with **low light** (<1% surface light).

It fixes ~4 Gt of carbon per year and contributes ~20% of O_2 to the atmosphere. Despite this, it was not discovered until 1985.



Chlorophyll-*a*:
$R^8 = C_2H_5$
Divinyl-chlorophyll-*a*:
$R^8 = CH{=}CH_2$

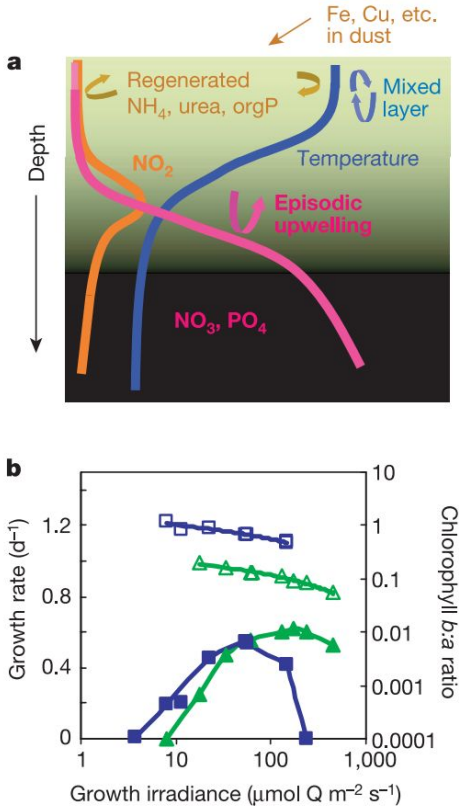


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10.1073/pnas.1307701110

Cyanobacteria: *Prochlorococcus marinus*

Prochlorococcus has different "ecotypes" occupying different niches, and can vary by pigments, light requirements, nitrogen, phosphorus and copper utilization, and virus sensitivity. They can be differentiated on the basis of the sequence of the ribosomal RNA gene, grouped into two different subspecies. **Low-light Adapted** (MIT9313 - blue) are most abundant at the bottom of the euphotic zone, where nutrient concentrations are high and light energy low. **High-light Adapted** (MED4 - green) are most abundant near the surface, adapted to strong light conditions and low nutrient concentrations.

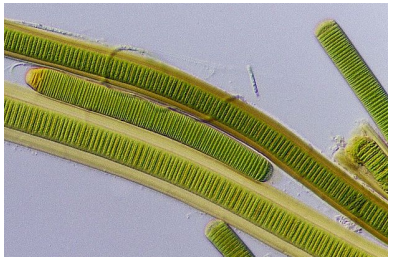


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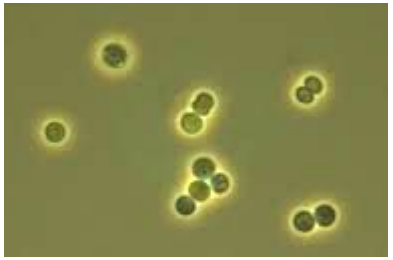
doi.org/10.1038/nature01947

Cyanobacteria: *Prochlorococcus marinus*

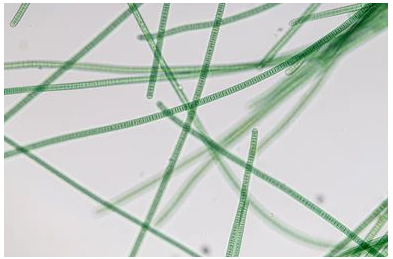
Lyngbya majuscula is a filamentous cyanobacterium commonly found in **benthic** habitats in both **coastal** and **estuarine** environments. Can form dense mats, often associated with **harmful algal blooms**, and some species produce toxins that can affect marine organisms and human health. It is known for producing secondary metabolites, including some with antimicrobial and cytotoxic properties.



Crocosphaera watsonii is a unicellular **nitrogen-fixing** cyanobacterium with very small size (2.5 - 3 µm in diameter). It's commonly found in tropical and subtropical oligotrophic waters, particularly in nutrient-poor areas where nitrogen is a limiting factor. Its nitrogen fixation is tightly **regulated by circadian rhythms**, with photosynthesis occurring during the day and nitrogen fixation at night.

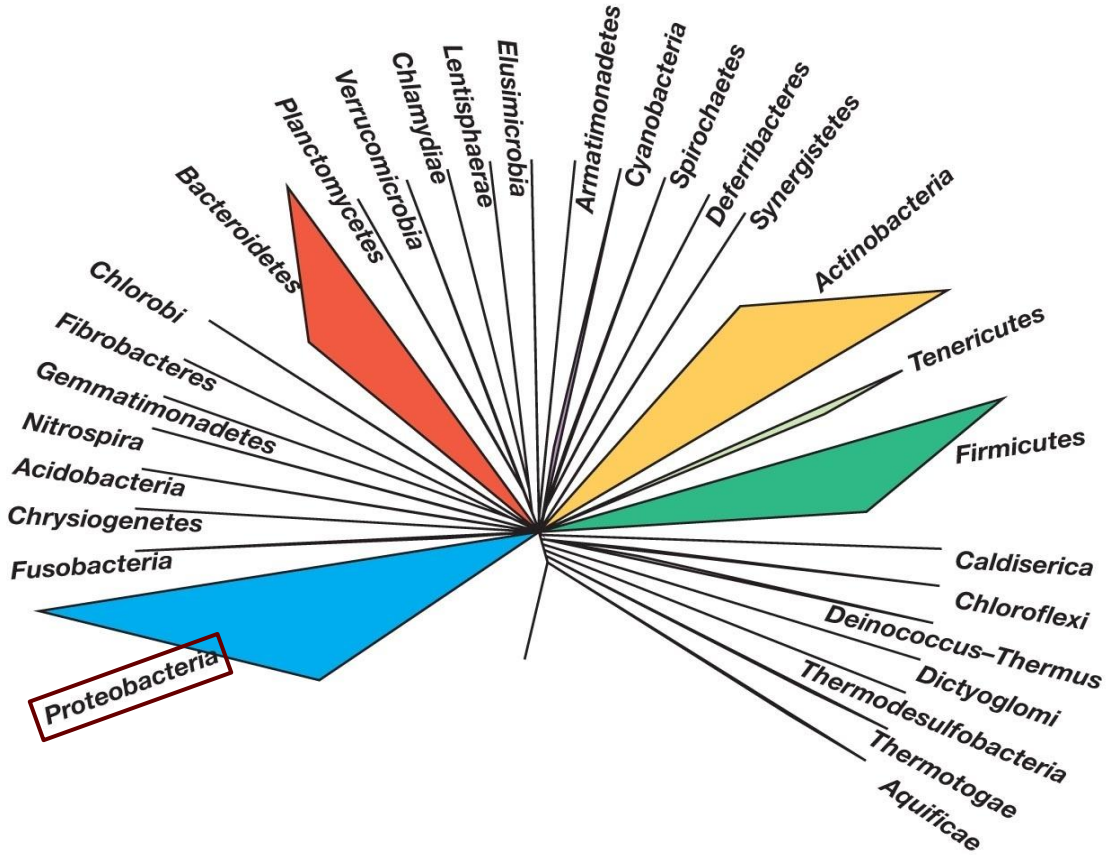


Oscillatoria brevis is a species of cyanobacteria belonging to the genus Oscillatoria, which is known for its filamentous morphology and gliding motility. **Found in** a variety of aquatic environments, especially **shallow waters**, near seagrass beds, **coastal sediments**, and **estuaries**. It can also thrive in hypersaline environments like salt marshes, tolerating changes in salinity and temperature.



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The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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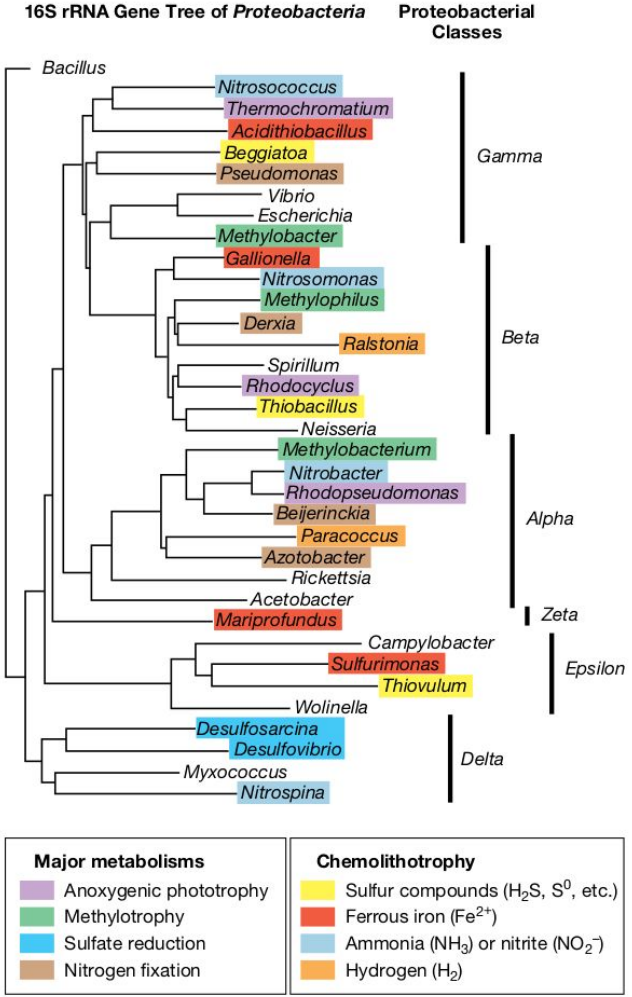
Bacterial Phyla: Proteobacteria

The Proteobacteria are a **major phylum** of Gram-negative bacteria. Carl Woese established this grouping in 1987, calling it informally the "purple bacteria and their relatives". Because of the great diversity of forms found in this group, the Proteobacteria are named after Proteus, a Greek god of the sea capable of assuming many different shapes

They are ubiquitous in the environments and posses a huge metabolic diversity, being facultative or obligate anaerobes, chemolithoautotrophs, heterotrophs and anoxygenic phototrophs. They include free living, commensal, symbiotic and pathogenic species.

They are currently divide in six classes: Alphaproteobacteria, Betaproteobacteria, Gammaproteobacteria, Deltaproteobacteria, Epsilonproteobacteria and Zetaproteobacteria

Given their diversity they have been proposed as superphylum, with the classes promoted to distinct phyla



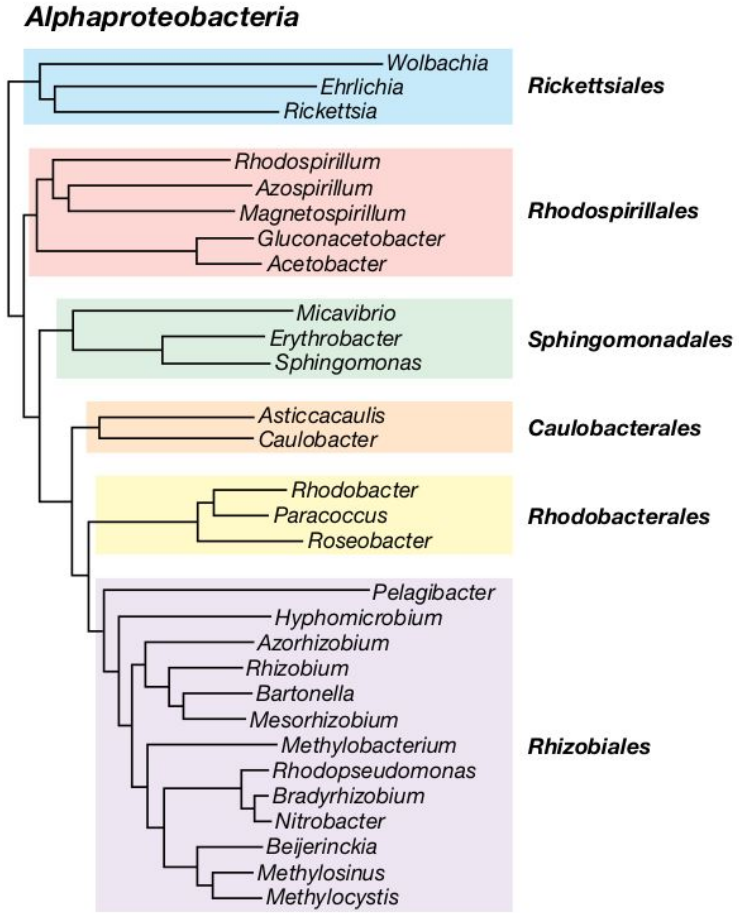
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Proteobacteria: Alphaproteobacteria

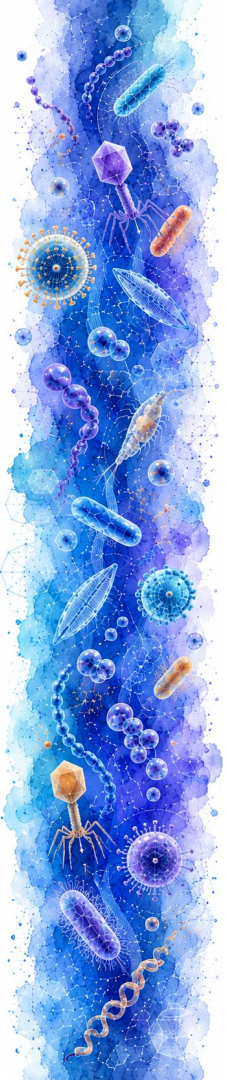
With nearly **1000 described species**, the Alphaproteobacteria are the largest class of Proteobacteria, comprising several phototrophs, methylotrophs, diazotrophic symbionts of plants and intracellular pathogens. A total of 10 orders have been described within the Alphaproteobacteria.

Most species are obligate aerobes or facultative aerobes and many are oligotrophic, preferring to grow in environments that have low nutrient concentration. Several species play a key role in nitrogen fixation in various types of plants.

The class represent the proto-mitochondrion, the hypothetical ancestral bacterial endosymbiont from which all mitochondria in eukaryotes are thought to descend, after an episode of symbiogenesis which created the aerobic eukaryotes.



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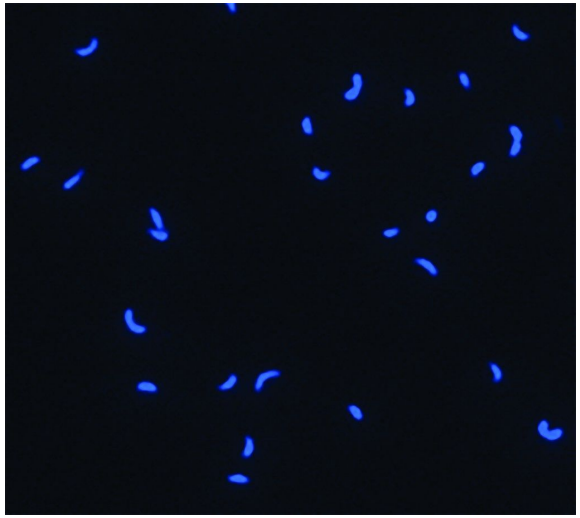
Proteobacteria: Alphaproteobacteria

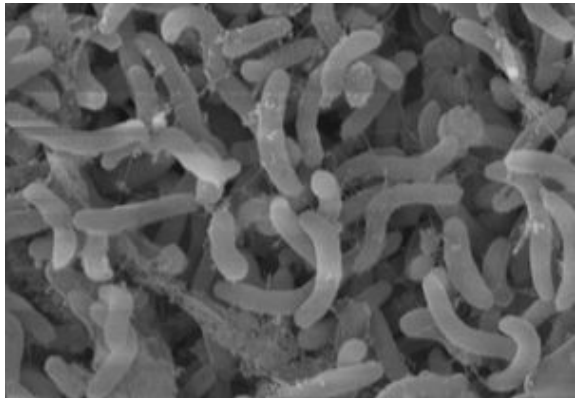
Pelagibacter ubique is an **oligotrophic** and an **obligately aerobic chemoorganotroph** that inhabits the photic zone, reaching the 25% of the bacterial cells found at the ocean's surface, and the 50% of cells in temperate waters in the summer. As a consequence, Pelagibacter ubique is likely the most abundant bacterial species belonging to Ca. Pelagibacter genus, with the single species *Ca. P. ubique* as an abundant member of the SAR11 clade in the phylum Alphaproteobacteria

It recycles dissolved organic carbon, thus playing a major role in the oceans carbon cycle.

It is rod or crescent shaped and one of the **smallest cells** known, with a length of 0.37-0.89 μm and a diameter of only 0.12-0.20 μm, with almost 30% of the cell's volume taken up by its genome.

A high-throughput method in microtiter plates containing sterile seawater supplemented with phosphate, ammonia, or low levels of organic compounds allowed the physiological properties of this ubiquitous organism to be investigated for the first time





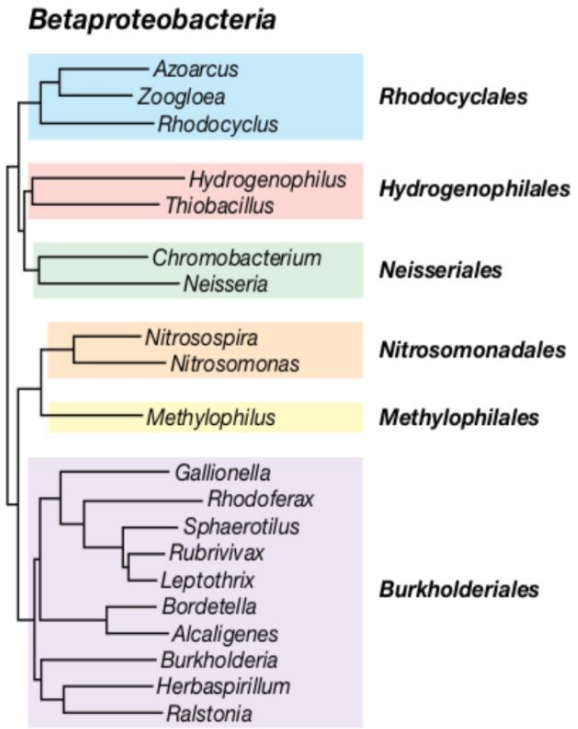
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Proteobacteria: Betaproteobacteria

With nearly **500 described species**, the Betaproteobacteria are the second largest class of Proteobacteria, but found only in low abundance in marine environments, mostly found in coastal and estuarine habitat.

The Betaproteobacteria contain species with a wide range of metabolic and ecological characteristics, including strictly aerobic, facultatively aerobic, and obligately anaerobic chemoorganotrophs, anoxygenic phototrophs, obligate and facultative chemolithotrophs.

The Methylophilales and Nitrosomonadales contain metabolically specialized organisms. Methylophilus species are obligate and facultative methylotrophs that grow on methanol and other C1 compounds, but not on CH_4. Whereas the order Nitrosomonadales contains obligately chemolithotrophic ammonia-oxidizing bacteria, the key genera being Nitrosomonas and Nitrosospira



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Betaproteobacteria: *Nitrosomonas marina*

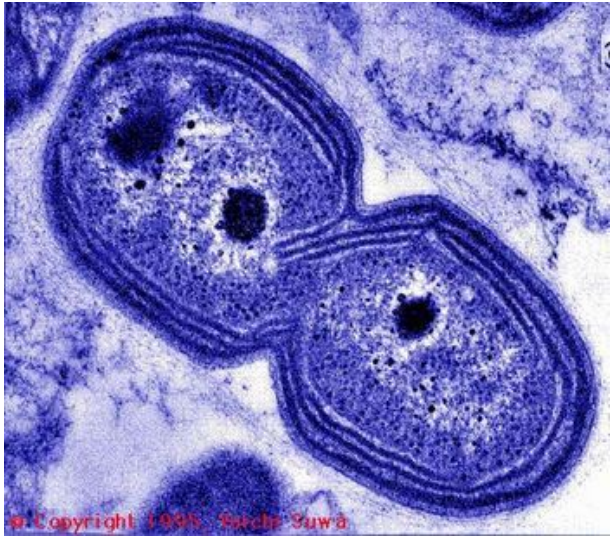
Nitrosomonas is a genus of rod-shaped chemolithoautotrophic bacteria in the phylum Betaproteobacteria

Nitrosomonas prefers an optimum pH of 6.0-9.0 and a temperature range of 20 to 30 °C. Most species are motile with a flagellum located in the polar regions

Nitrosomonas species are aerobic chemolithoautotrophs using the Calvin cycle and oxidize NH_3 to NO_2^-, and is therefore known as ammonia oxidizer.

Same Nitrosomonas species have been identified as also being able to degrade a variety of halogenated compounds including trichloroethylene, benzene, and vinyl chloride

Together with nitrifying bacteria, aerobic ammonia oxidizers can be used as biofilters in aquaculture systems, since they convert toxic ammonia to less toxic nitrite and nitrate.



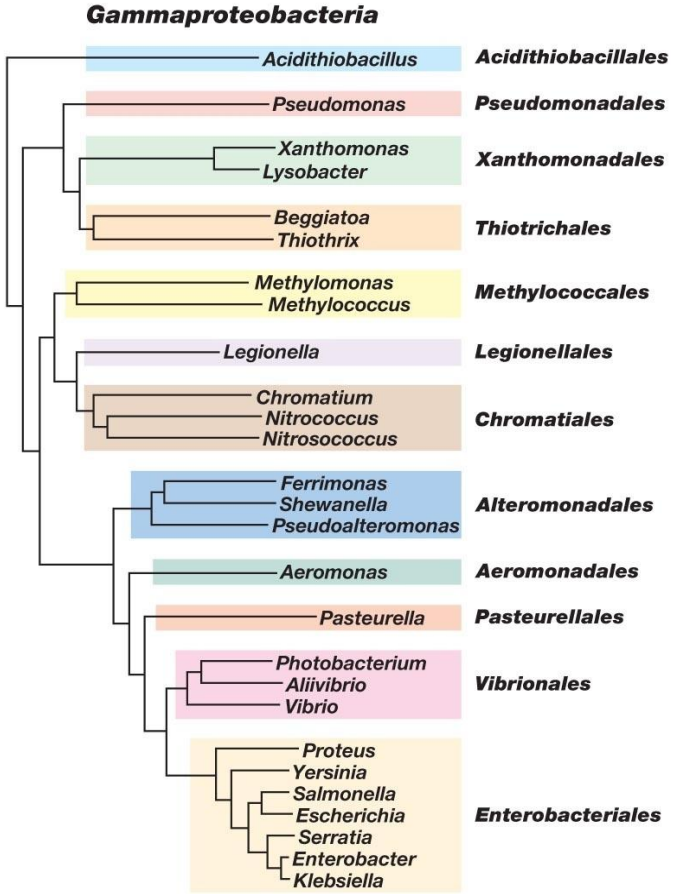
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Proteobacteria: Gammaproteobacteria

Gammaproteobacteria are major players in diverse marine ecosystems, including extreme environments, and comprise several medically, ecologically and scientifically important groups of bacteria.

The class contains a huge variety of taxonomic and metabolic diversity, including aerobic and anaerobic species, chemolithoautotrophic, chemoorganotrophic and phototrophic species and include free living, biofilms formers, commensal and symbionts

Chemolithoautotrophic species use the Calvin cycle to fix CO_2, with the exception of the symbionts of the vent tubeworm Riftia which posses also the rTCA cycle
Families of interest include the Thiotrichales and Chromatiales, Pseudomonadales, Vibrionales, Methylococcales and Enterobacterales

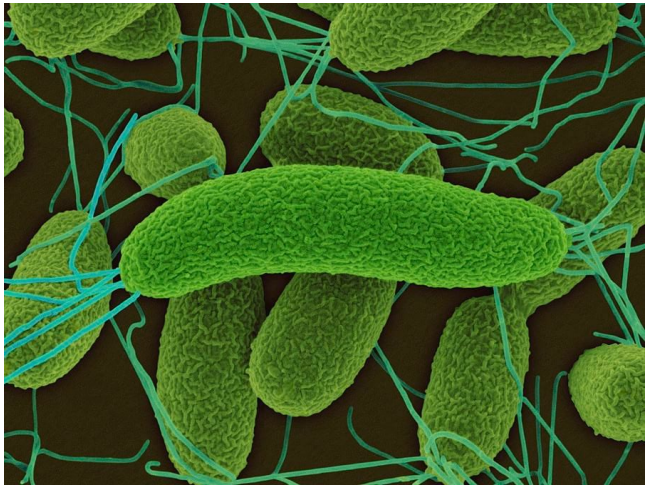


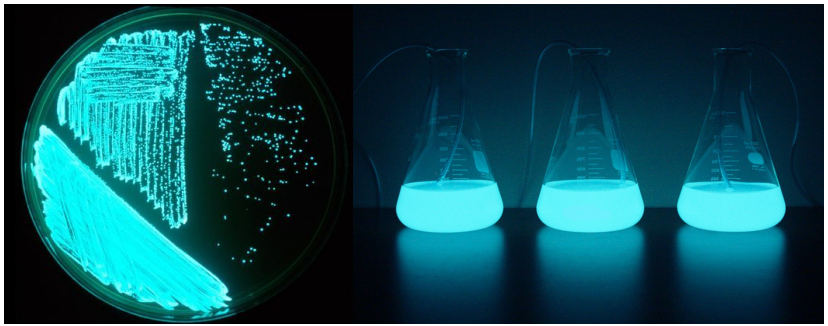
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Gammaproteobacteria: *Vibrio fischeri*

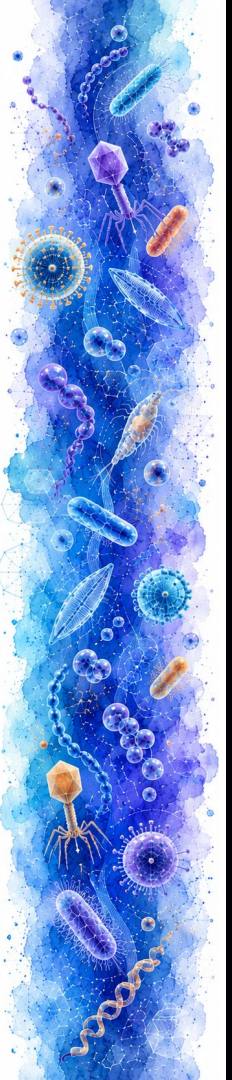
Vibrio fischeri is a species of Gram-negative, **chemoheterotrophic**, primarily utilizing organic compounds as carbon and energy sources. Capable of both **aerobic respiration** and **fermentation**, depending on the availability of oxygen.

Vibrio fischeri is also a bioluminescent bacteria that is commonly found in marine environments. The bioluminescence of *V. fischeri* is controlled by **quorum sensing**, a mechanism that bacteria use to communicate with each other through the production and detection of signaling molecules called autoinducers.





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Euprymna scolopes

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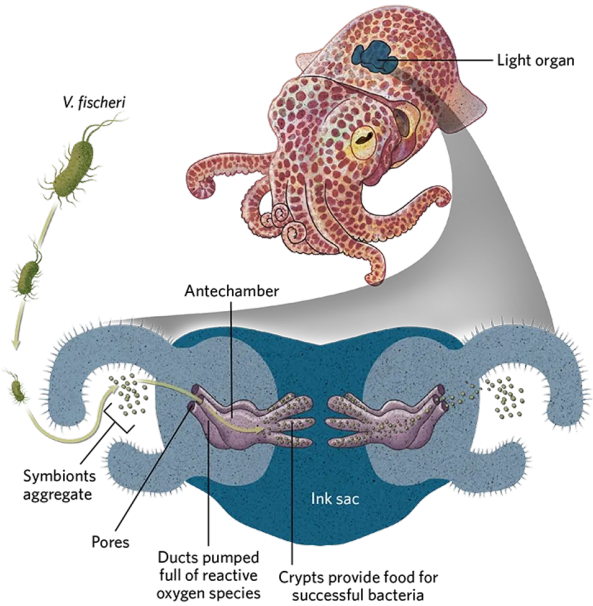
Todd Bretl Underwater Photography

Symbiotic Relationships

V. fischeri bioluminescence capacity is particularly known for its **mutualistic symbiotic relationships** with various marine organisms, most notably the Hawaiian bobtail squid *Euprymna scolopes*.

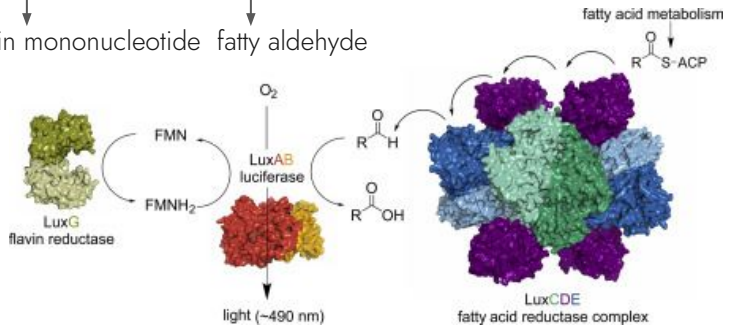
Squid advantage: camouflage strategy to reduce predation (+)

Bacterial advantage: get nutrients and a secure environment for thriving (+)



$$FMNH_2 + O_2 + R\text{-}CHO \xrightarrow{\text{Luciferase}} FMN + H_2O + R\text{-}COOH + Light$$

flavin mononucleotide fatty aldehyde



1. Mucus Production and Ciliary Sweeping
2. Acidic Microenvironment (V-type ATPase)
3. Chemical and Immune Defense
4. Specific Molecular Recognition (exopolysaccharide + outer-membrane proteins)

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https://doi.org/10.1038/s41579-021-00557-0

Gammaproteobacteria: *Vibrio parahaemolyticus*

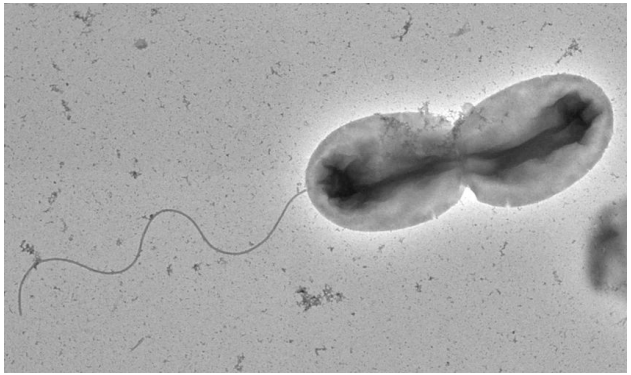
Vibrio parahaemolyticus is a curved, rod-shaped member of the *Vibrionales*

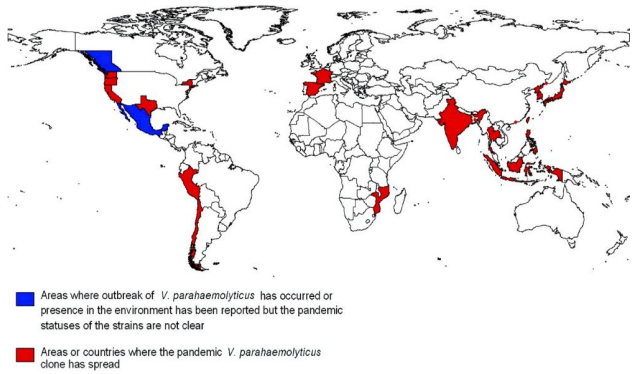
It is found in brackish saltwater, and generally associated with shellfish

V. parahaemolyticus is facultatively aerobic, motile, and does not form spores

It is the predominant cause of acute gastroenteritis generally caused by the ingestion of bacteria in raw or undercooked seafood, usually oysters, squid, mackerel, tuna, crabs

Outbreaks tend to be concentrated along coastal regions during the summer and early fall when higher water temperatures favor higher levels of bacterial growth.





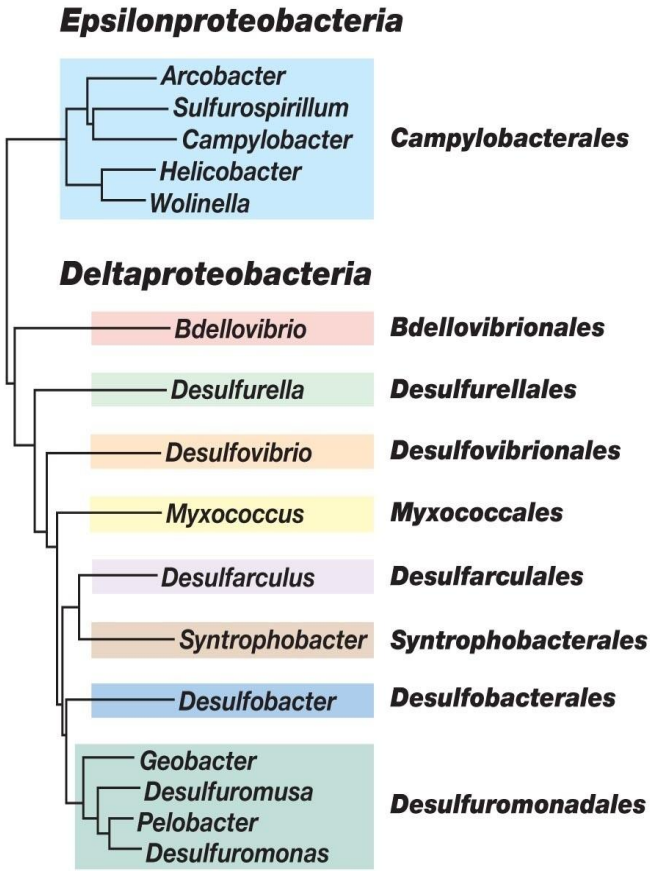
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Proteobacteria: Deltaproteobacteria

The Deltaproteobacteria are ubiquitous in marine sediments, and contains most of the known sulfate- and sulfur-reducing bacteria, alongside several other anaerobic bacteria with different metabolic lifestyles, such as the ferric iron-reducing Geobacter *spp.*

The largest and most common order containing sulfate reducers is the Desulfovibrionales, followed by the Desulfobacterales and Desulfarculales, isolated from marine sediments, especially if nutrient-rich anoxic environments that contain sulfate

Sulfate reduction leads to a release of H_2S in the marine environment, resulting in the characteristic stench and blackening of decomposition in anoxic mud, sediments, and decaying seaweed, due to deposits of FeS. H_2S is highly toxic and adversely affects many forms of marine life, but it can be used by a wide range of chemotrophic and phototrophic bacteria, completing the cycling of sulfur.



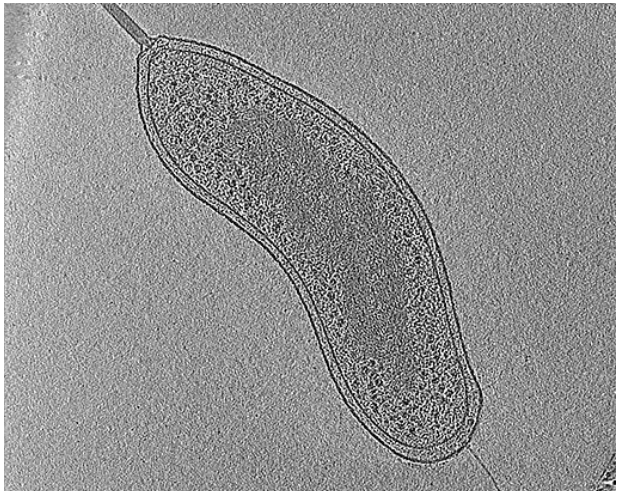
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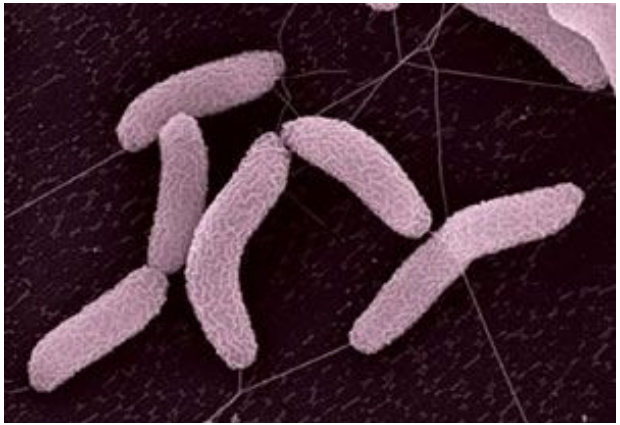
Proteobacteria: Deltaproteobacteria

Species of *Desulfovibrionales* are **obligately anaerobic sulfate reducers**, using **sulfate** as the **terminal electron acceptor**, and all **require small organic compounds**, such as lactate, as a **source of carbon** and **energy** for growth.

Species within the orders *Desulfobacterales* and *Desulfarculales* can be complete or incomplete acetate oxidizers. In addition to sulfate, some species in these three orders can also reduce sulfite, thiosulfate, or nitrate, and some are capable of certain fermentations. Some species are able to fix CO_2 using the reductive Acetyl-CoA pathway (WL) or rTCA cycle

Some but not all *Syntrophobacterales* are able to reduce sulfate. In nature, however, species of ***Syntrophobacterales*** **primarily interact with hydrogenotrophic bacteria** in a metabolic partnership called syntrophy, **where electrons coming from hydrogen oxidation**, performed by H_2 oxidizers, **are used by *Syntrophobacterales* for sulfate reduction**.





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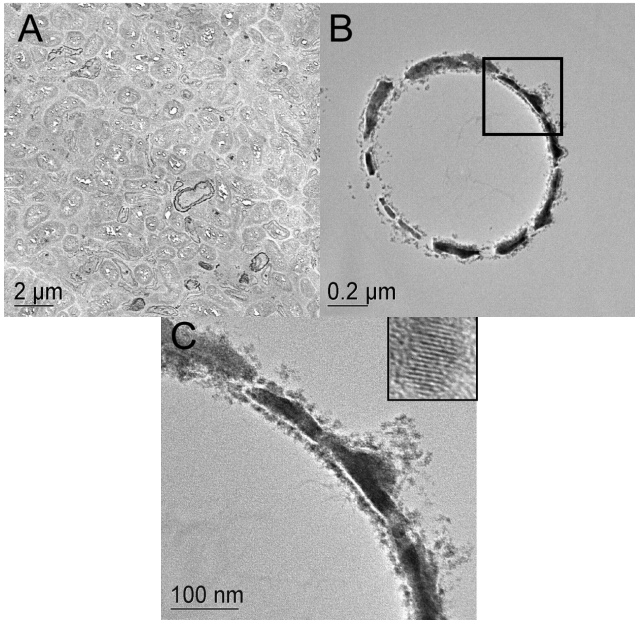
Deltaproteobacteria: *Desulfovibrio desulfuricans*

Desulfovibrio desulfuricans is a motile, anaerobic sulfate-reducing bacterium, generally found in marine or freshwater sediments, and the guts of ruminants, and, although in rare cases, it has been found to cause infection in humans, causing primary bacteremia and abdominal infections, such as abscess and cholecystitis

Desulfovibrio cells typically grow anaerobically, but certain strains have been identified growing in the presence of oxygen. It measures approximately 3 μm by 0.5 μm

Desulfovibrio is known for its metabolic flexibility in response to the extended amount of electron acceptors it utilizes including sulfate, sulfur, nitrate, and nitrite among others

Species of Desulfovibrio were first identified as possible bioremediator, thanks to its ability to reduce several toxic metals such as uranium (VI) and chromium (VI)



Desulfovibrio biofilm cells with mineral precipitation in the periplasm and in the extracellular polymeric matrix. Clearly visible well-defined nanocrystalline character of the precipitated uranium, with a crystal size of 2–5 nm.

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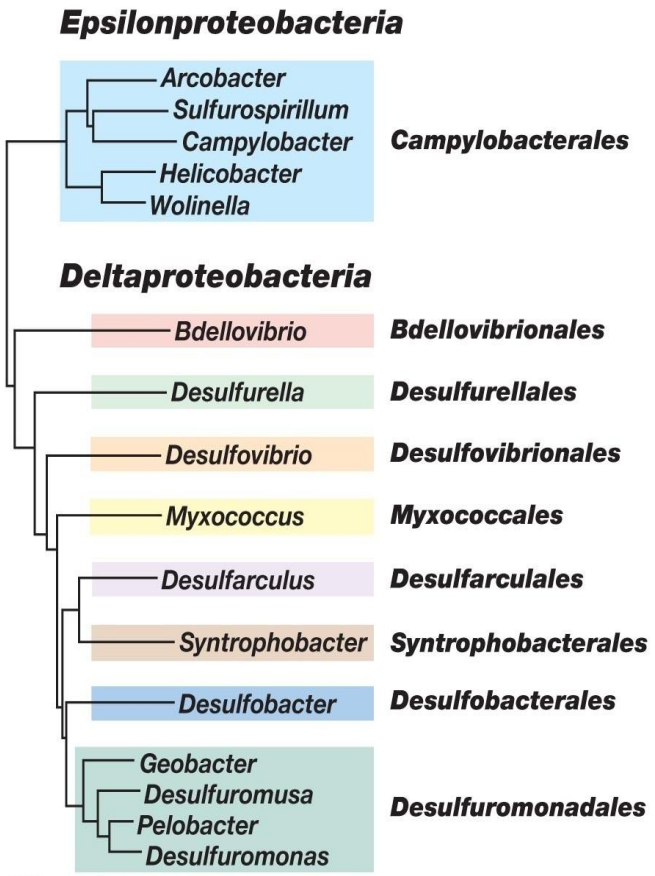
10.1021/es071335k

Proteobacteria: Epsilonproteobacteria

The Epsilonproteobacteria consist of few known genera, including free-living or symbionts species found in sulfidic environments, and some mammalian commensal or pathogenic species

Epsilonproteobacteria are major player in sulfidic marine environments. Numerous environmental sequences and isolates of Epsilonproteobacteria have been recovered from hydrothermal vents and cold seep habitats. Examples of isolates include *Sulfurimonas autotrophica*, *Sulfurovum lithotrophicum* and *Caminibacter mediatlanticus*

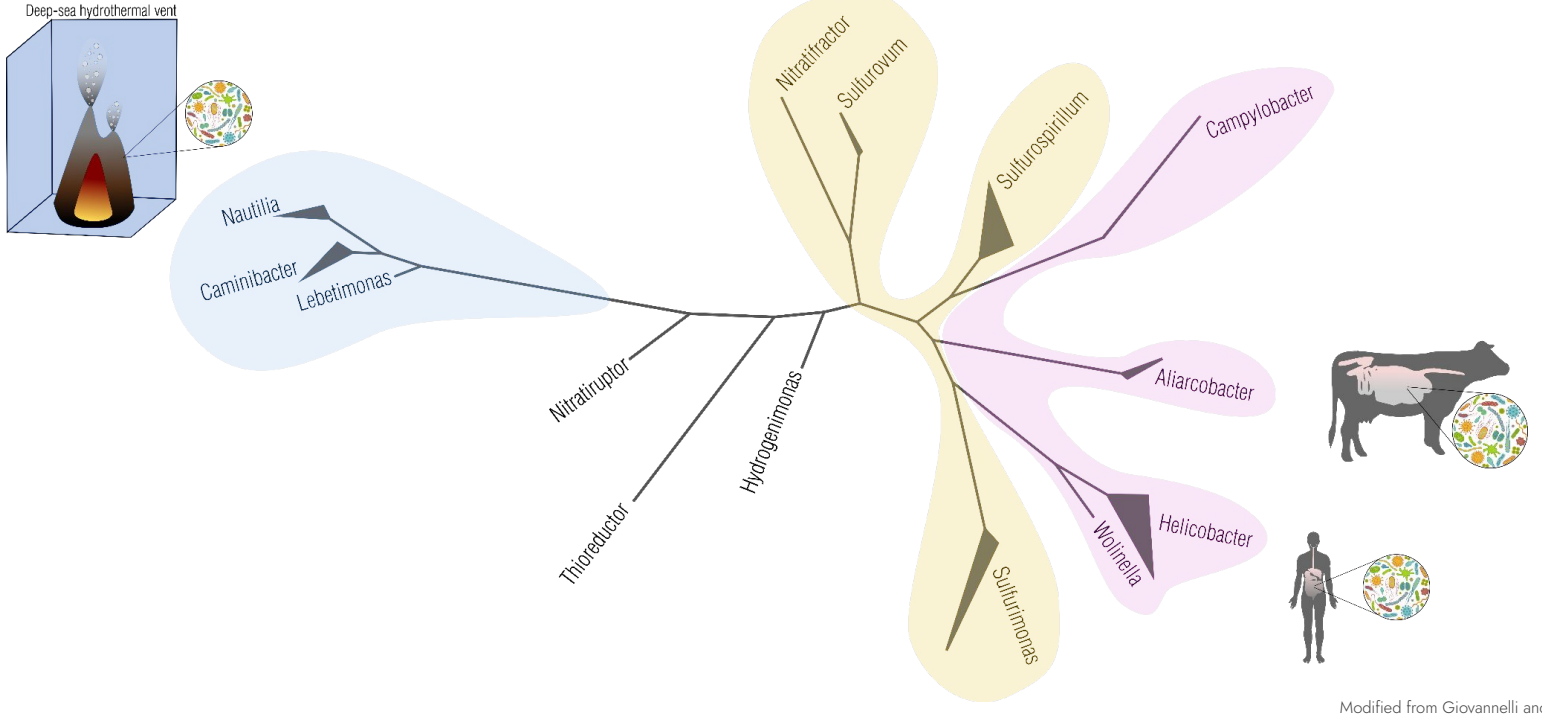
The Epsilonproteobacteria found at deep-sea hydrothermal vents are typically **anaerobic or microaerophilic chemolithotrophs**, and **oxidize reduced sulfur, formate, or hydrogen coupled to the reduction of nitrate or oxygen**. They use the **rTCA cycle to fix carbon dioxide into biomass**, a pathway originally thought to be of little environmental significance



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Proteobacteria: Epsilonproteobacteria

Deep-sea vents → Sulfidic → Commensal/Pathogenic



Modified from Giovannelli and Vetriani (2017)

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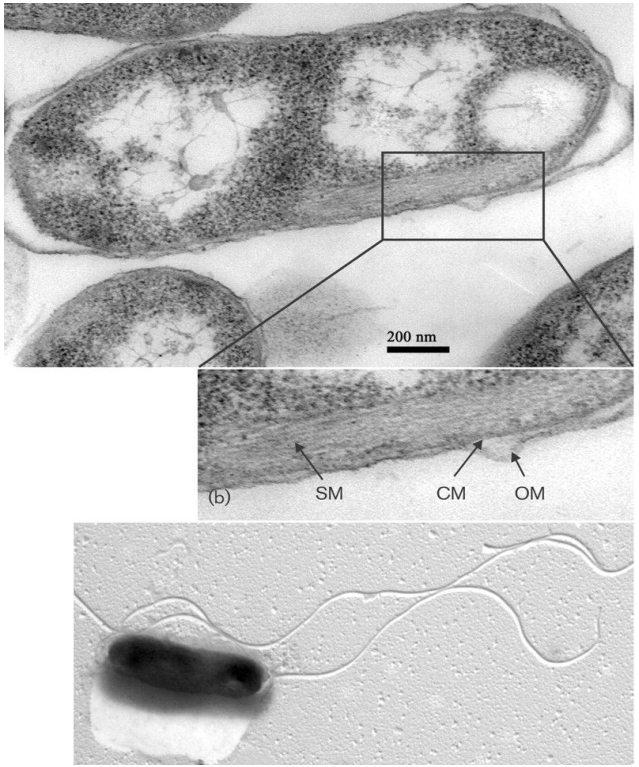
Epsilonproteobacteria: *Caminibacter mediatlanticus*

Epsilonproteobacteria were initially defined by only a few pathogenic species belonging to Campylobacter and Helicobacter genus

Lately discovered environmental species, are anaerobic, chemolithoautotrophic thermophiles, especially abundant in anoxic sulfur-rich environments

They are hydrogenotrophs, using H_2 as the energy source and CO_2 as the carbon source, whereas nitrate or elemental sulfur are used as the electron acceptor, with resulting production of ammonium and hydrogen sulfide

C. mediatlanticus was isolated from the walls of an active deep-sea hydrothermal vent chimney on the Mid-Atlantic Ridge, and usually grows at temperatures ranges of 45-70°C, with an optimum at 55°C. *C. mediatlanticus* is endowed with peculiar stacked membranes probably evolved for a better organization of enzymes of complex metabolic reactions



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Proteobacteria: Zetaproteobacteria

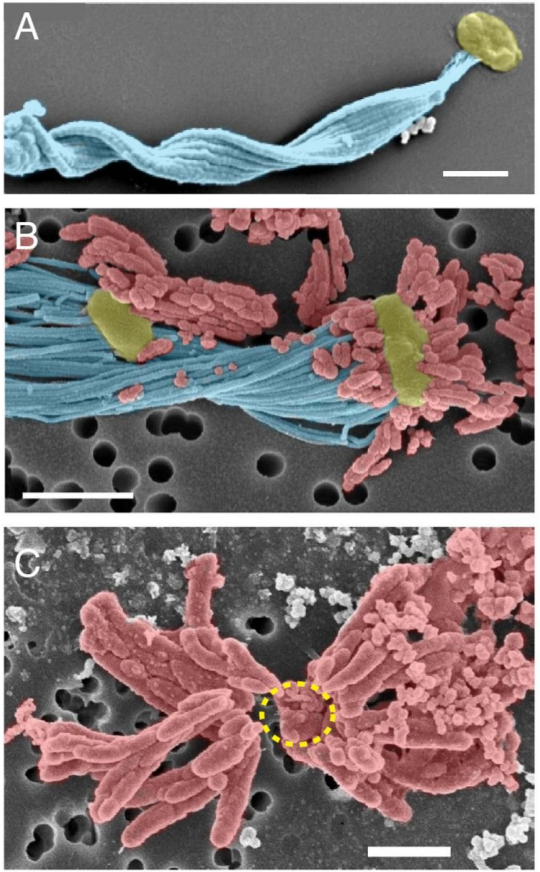
Zetaproteobacteria are a new class within the Proteobacteria consisting of a single cultivated genus Mariprofundus

Zetaproteobacteria are now believed to be key player in the marine iron biogeochemical cycle

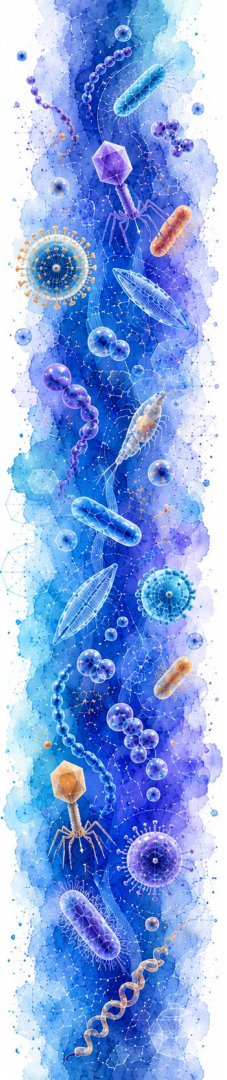
Mariprofundus ferrooxydans was first isolated from Lō'ihi Seamount, a submarine volcano found near Hawaii

M. ferrooxydans is a neutrophilic aerobic iron-oxidizer, forming twisted stalk- or short dread-like structures containing extracellular organo-metallic structures containing Fe(III) oxyhydroxides, obtained from the oxidation of ferrous iron. The iron-encrusted stalks accumulate, as it is excreted from the cell surface. Stalk formation is presumably an adaptation that prevents cells from becoming buried in a crust of iron oxide

M. ferrooxydans is a autotrophic chemolithotrophs fixing CO_2 using the Calvin cycle



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This Week Read

Salazar G, Sunagawa S. Marine microbial diversity. Curr Biol. 2017;27(11):R489-R494. doi:10.1016/j.cub.2017.01.017

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